

```

import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import numpy as np
from google.colab import drive
drive.mount('/content/drive')
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import plotly.express as px
import shutil

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```

numeric_columns = [
    'poids_de_la_discipline', 'taux_dinsertion',
    'emplois_cadre_ou_professions_intermediaires',
    'emplois_stables', 'emplois_a_temps_plein',
    'salaire_net_median_des_emplois_a_temps_plein',
    'salaire_brut_annuel_estime', 'de_diplomes_boursiers',
    'emplois_cadre',
    'emplois_exterieurs_a_la_region_de_luniversite', 'annee',
    'salaire_net_mensuel_median_regional', 'femmes'
]

```

```

df = pd.read_csv('/content/fr-esr-insertion_professionnelle-
master.csv', sep=';')

```

```

for col in numeric_columns:
    df[col] = pd.to_numeric(df[col], errors='coerce')

```

```
df
```

```
{"type": "dataframe", "variable_name": "df"}
```

```
df.tail()
```

```
{"type": "dataframe"}
```

```
df.shape
```

```
(19603, 35)
```

Data cleaning

```
df.isna().sum(axis=0)
```

annee	0
diplome	0
numero_de_l_etablissement	0
etablissement	0
etablissementactuel	15149
code_de_l_academie	361
academie	361
code_du_domaine	0
domaine	0
code_de_la_discipline	0
discipline	0
situation	0
remarque	18034
nombre_de_reponses	47
taux_de_reponse	321
poids_de_la_discipline	838
taux_dinsertion	8747
taux_d_emploi	17669
taux_d_emploi_salarie_en_france	17669
emplois_cadre_ou_professions_intermediaires	9787
emplois_stables	9668
emplois_a_temps_plein	9734
salaire_net_median_des_emplois_a_temps_plein	11341
salaire_brut_annuel_estime	11161
de_diplomes_boursiers	198
taux_de_chomage_regional	0
salaire_net_mensuel_median_regional	682
emplois_cadre	10262
emplois_exterieurs_a_la_region_de_luniversite	10067
femmes	9245
salaire_net_mensuel_regional_1er_quartile	2025
salaire_net_mensuel_regional_3eme_quartile	2025
cle_etab	0
cle_disc	0
id_paysage	361
dtype: int64	

```
df.isna().sum(axis=0).sum().sum()
```

```
165752
```

```
round(df.isna().sum(axis=0) / df.shape[0] * 100)
```

annee	0.0
diplome	0.0
numero_de_l_etablissement	0.0
etablissement	0.0
etablissementactuel	77.0
code_de_l_academie	2.0
academie	2.0

code_du_domaine	0.0
domaine	0.0
code_de_la_discipline	0.0
discipline	0.0
situation	0.0
remarque	92.0
nombre_de_reponses	0.0
taux_de_reponse	2.0
poids_de_la_discipline	4.0
taux_dinsertion	45.0
taux_d_emploi	90.0
taux_d_emploi_salarie_en_france	90.0
emplois_cadre_ou_professions_intermediaires	50.0
emplois_stables	49.0
emplois_a_temps_plein	50.0
salaire_net_median_des_emplois_a_temps_plein	58.0
salaire_brut_annuel_estime	57.0
de_diplomes_boursiers	1.0
taux_de_chomage_regional	0.0
salaire_net_mensuel_median_regional	3.0
emplois_cadre	52.0
emplois_exterieurs_a_la_region_de_luniversite	51.0
femmes	47.0
salaire_net_mensuel_regional_1er_quartile	10.0
salaire_net_mensuel_regional_3eme_quartile	10.0
cle_etab	0.0
cle_disc	0.0
id_paysage	2.0
dtype:	float64

```
print(df.dtypes)
```

annee	int64
diplome	object
numero_de_l_etablissement	object
etablissement	object
etablissementactuel	object
code_de_l_academie	object
academie	object
code_du_domaine	object
domaine	object
code_de_la_discipline	object
discipline	object
situation	object
remarque	object
nombre_de_reponses	float64
taux_de_reponse	object
poids_de_la_discipline	float64
taux_dinsertion	float64
taux_d_emploi	object

taux_d_emploi_salarie_en_france	object
emplois_cadre_ou_professions_intermediaires	float64
emplois_stables	float64
emplois_a_temps_plein	float64
salaire_net_median_des_emplois_a_temps_plein	float64
salaire_brut_annuel_estime	float64
de_diplomes_boursiers	float64
taux_de_chomage_regional	object
salaire_net_mensuel_median_regional	float64
emplois_cadre	float64
emplois_exterieurs_a_la_region_de_luniversite	float64
femmes	float64
salaire_net_mensuel_regional_1er_quartile	object
salaire_net_mensuel_regional_3eme_quartile	object
cle_etab	object
cle_disc	object
id_paysage	object
dtype:	object

```
for col in df.select_dtypes(include='object').columns:
    unique_values = df[col].unique()
    print(f"Valeurs uniques pour '{col}':\n{unique_values}\n")
```

Valeurs uniques pour 'diplome':
 ['MASTER LMD' 'MASTER ENS']

Valeurs uniques pour 'numero_de_l_etablissement':

```
['0751719L' '0755890V' '0755976N' '0761904G' '0762762P' '0772502B'
'0781944P' '0801344B' '0860856N' '0870669E' '0900424X' '0912330N'
'0921204J' '0931238R' '0931827F' '0941111X' '0951793H' '7200664J'
'9710585J' '9730429D' '9740478B' '9840349G' 'UNIV' '0060931E'
'0101060Y'
'0134009M' '0141408E' '0171463Y' '0211237F' '0251215K' '0251985X'
'0290346U' '0301687W' '0311382J' '0311384L' '0333298F' '0341089Z'
'0342321N' '0350936C' '0350937D' '0383493R' '0421095M' '0450855K'
'0490970N' '0511296G' '0542493S' '0561718N' '0593279U' '0595964M'
'0597065J' '0623957P' '0632035V' '0640251A' '0660437S' '0673021V'
'0681166Y' '0691774D' '0691775E' '0692437Z' '0720916E' '0730858L'
'0751717J' '0830766G' '0911975C' '0062205P' '0311383K' '0331766R'
'0370800U' '0383546Y' '0442953W' '0597132G' '0632084Y' '0751718K'
'0772894C' '0811293R' '0840685N' '0912408Y' '0952259P' '0342490X'
'0751720M' '0751721N' '0751723R' '0911101C' '0440984F' '0593559Y'
'0593560Z' '0593561A' '0751722P' '0631525R' '0750736T' '0631262E'
'0331764N' '0341087X' '0341088Y' '0381838S' '0381839T' '0381840U'
'0756305W' '0332929E' '0331765P' '0596870X']
```

Valeurs uniques pour 'etablissement':
 ['Paris 3 - Sorbonne Nouvelle' 'Sorbonne Université' 'Université de Paris'
 'Rouen' 'Le Havre' 'Paris-Est Marne-la-Vallée']

'Versailles Saint-Quentin-en-Yvelines' 'Amiens - Picardie Jules-Verne'
 'Poitiers' 'Limoges' 'Université de technologie de Belfort-Montbéliard'
 'Paris-Saclay' 'Paris Nanterre' 'Université Sorbonne Paris Nord'
 'Paris 8 - Vincennes Saint-Denis' 'Paris-Est Créteil' 'Cergy-Pontoise'
 'Corse Pasquale Paoli' 'Antilles' 'Guyane' 'La Réunion'
 'Polynésie Française' 'Toutes universités et établissements assimilés'
 'Nice - Sophia-Antipolis' 'Université de technologie de Troyes'
 'Aix-Marseille' 'Caen Normandie' 'La Rochelle' 'Dijon - Bourgogne'
 'Besançon - Franche-Comté' 'COMUE UBFC' 'Brest - Bretagne occidentale'
 'Nîmes' 'Toulouse 1 - Capitole' 'Toulouse 3 - Paul Sabatier'
 'Bordeaux'
 'Montpellier 3 - Paul-Valéry' 'Montpellier' 'Rennes 1' 'Rennes 2'
 'Grenoble Alpes' 'Saint-Etienne - Jean Monnet' 'Orléans' 'Angers'
 'Reims - Champagne Ardenne' 'Lorraine' 'Bretagne-Sud'
 'Université Polytechnique Hauts-de-France' "Littoral - Côte d'Opale"
 'Lille' 'Artois' 'Clermont Auvergne' "Pau - Pays de l'Adour"
 'Perpignan - Via Domitia' 'Strasbourg' 'Mulhouse - Haute Alsace'
 'Lyon 1 - Claude Bernard' 'Lyon 2 - Lumière' 'Lyon 3 - Jean Moulin'
 'Le Mans' 'Savoie Mont Blanc' 'Paris 1 - Panthéon Sorbonne' 'Toulon'
 "Évry-Val d'Essonne" "Côte d'Azur" 'Toulouse - Jean Jaures'
 'Bordeaux - Montaigne' 'Tours' 'Nantes' 'Polytechnique Hauts-de-France'
 'Paris 2 - Panthéon Assas' 'Gustave Eiffel'
 'INU Jean-François Champollion' 'Avignon' 'CY Cergy Paris'
 'Paris 4 - Sorbonne' 'Paris René Descartes' 'Paris 7 - Denis Diderot'
 'Paris 11 - Paris Sud' 'Lille 1 - Sciences technologies'
 'Lille 2 - Droit et santé' 'Lille 3 - Charles-de-Gaulle'
 'Paris Pierre et Marie Curie' 'Clermont-Ferrand 2 - Blaise Pascal'
 'Paris - Dauphine' 'Clermont-Ferrand 1 - Auvergne'
 'Bordeaux 1 - Sciences technologies' 'Montpellier 1'
 'Montpellier 2 - Sciences techniques du Languedoc'
 'Grenoble 1 - Joseph Fourier' 'Grenoble 2 - Pierre Mendès France'
 'Grenoble 3 - Stendhal' 'Paris-Panthéon-Assas' 'Bordeaux 4 - Montesquieu'
 'Bordeaux 2 - Victor Segalen' 'Lille Nord de France']

Valeurs uniques pour 'etablissementactuel':

[nan 'Université Gustave Eiffel (UGE)' 'Université Paris-Saclay'
 'CY Cergy Paris Université' "Université Côte d'Azur (UCA)"
 'Université Toulouse Capitole' 'Université de Montpellier'
 'Université de Rennes' 'Université Grenoble Alpes (UGA)'
 'Université Polytechnique Hauts-de-France (UPHF)' 'Université de Lille'
 'Université Clermont Auvergne (UCA)' 'Nantes Université'

'Université Paris-Panthéon-Assas' 'Sorbonne Université (Sorbonne U)'
'Université Paris Cité' 'Université de Bordeaux']

Valeurs uniques pour 'code_de_l_academie':

['A01' 'A70' 'A24' 'A25' 'A20' 'A13' 'A22' 'A03' 'A27' 'A32' 'A33'
'A28'
'A41' nan 'A23' 'A19' 'A02' 'A07' 'A14' 'A11' 'A16' 'A04' 'A08' 'A10'
'A18' 'A17' 'A12' 'A09' 'A06' 'A15']

Valeurs uniques pour 'academie':

['Paris' 'Normandie' 'Créteil' 'Versailles' 'Amiens' 'Poitiers'
'Limoges'
'Besançon' 'Corse' 'Guadeloupe' 'Guyane' 'La Réunion'
'Polynésie Française' nan 'Nice' 'Reims' 'Aix-Marseille' 'Dijon'
'Rennes'
'Montpellier' 'Toulouse' 'Bordeaux' 'Grenoble' 'Lyon' 'Orléans-Tours'
'Nantes' 'Nancy-Metz' 'Lille' 'Clermont-Ferrand' 'Strasbourg']

Valeurs uniques pour 'code_du_domaine':

['LLA' 'MEEF' 'DEG' 'STS' 'SHS']

Valeurs uniques pour 'domaine':

['Lettres, langues, arts' 'Masters enseignement'
'Droit, économie et gestion' 'Sciences, technologies et santé'
'Sciences humaines et sociales']

Valeurs uniques pour 'code_de_la_discipline':

['disc06' 'disc18' 'disc01' 'disc16' 'disc17' 'disc12' 'disc13'
'disc03'
'disc04' 'disc08' 'disc15' 'disc02' 'disc14' 'disc09' 'disc05'
'disc07'
'disc11' 'disc10' 'disc19' 'disc20']

Valeurs uniques pour 'discipline':

['Lettres, langues, arts' 'Masters enseignement'
'Ensemble formations juridiques, économiques et de gestion'
'Informatique' 'Autres sciences, technologies et santé'
'Ensemble sciences, technologies et santé'
'Sciences de la vie et de la terre' 'Économie' 'Gestion'
'Histoire-géographie' "Sciences de l'ingénieur" 'Droit'
'Sciences fondamentales' 'Psychologie'
'Autres formations juridiques, économiques et de gestion'
'Ensemble sciences humaines et sociales'
'Autres sciences humaines et sociales' 'Information communication'
'Masters enseignement : premier degré'
'Masters enseignement : second degré, CPE...']

Valeurs uniques pour 'situation':

['18 mois après le diplôme' '30 mois après le diplôme']

Valeurs uniques pour 'remarque':

```
[nan  
"Université ayant participé à l'enquête mais dont l'effectif est trop  
petit pour que les résultats soient jugés exploitables"  
'Université dont les résultats sont plus fragiles (taux de réponse  
inférieur à 50%)'  
"Université ayant participé à l'enquête mais dont le taux de réponse  
obtenu est insuffisant pour que les résultats soient jugés  
exploitables"  
"Université n'ayant pas participé à l'enquête"]
```

Valeurs uniques pour 'taux_de_reponse':

```
['61' '53' '62' '64' '59' '68' '71' '70' '69' '83' '80' '72' '73' '74'  
'42' '57' '60' '75' '78' '84' '81' '79' '47' '32' '50' '67' '66' '46'  
'43' '63' '56' '58' '52' '28' '45' '36' '39' '33' '76' '65' '77' nan  
'100' '25' '82' '85' '87' '89' '86' '90' '88' '54' '93' '48' '92'  
'91'  
'94' '95' '97' '44' '49' '20' '23' '41' '51' '55' '38' '40' '29' '35'  
'96' '31' '27' '26' 'ns' '37' '34' '0' '21' '22' '98' '30' '14' '18'  
'2'  
'24' '19' '8' '12' '6' '13' '15']
```

Valeurs uniques pour 'taux_d_emploi':

```
[nan '97' '77' 'ns' '92' '88' '86' '89' '95' '91' '94' '90' '82' '96'  
'81'  
'87' '84' '93' '75' '79' '80' '85' '74' '72' '62' '78' '100' '83'  
'98'  
'70' '76' '99' '67' '66' 'nd' '71' '73' '69' '61' '64' '54']
```

Valeurs uniques pour 'taux_d_emploi_salarie_en_france':

```
[nan '96' '63' 'ns' '81' '84' '77' '86' '90' '88' '78' '70' '83' '73'  
'79'  
'93' '62' '67' '80' '85' '72' '89' '82' '76' '69' '65' '57' '75' '87'  
'91' '71' '66' '47' '42' '92' '61' '98' '68' '59' '60' '74' '58' 'nd'  
'94' '97' '64' '95' '35' '55' '53' '46' '52' '100' '54' '51' '99'  
'36'  
'41']
```

Valeurs uniques pour 'taux_de_chomage_regional':

```
['7.1' '8' '10' '7.4' '7.8' '19.7' '18' '20.1' 'nd' '9' '7.7' '7.2'  
'6.6'  
'6.5' '9.3' '7' '6.7' '9.4' '19.4' '17.8' '17.4' '.' '8.3' '7.3'  
'6.9'  
'6.2' '5.8' '8.8' '6.4' '6' '8.9' '15.5' '19.2' '6.3' '8.6' '6.1'  
'5.7'  
'8.7' '11.9' '10.1' '10.2' '12.4' '11.4' '9.6' '10.8' '9.9' '11'  
'8.4'  
'8.1' '11.2' '8.2' '11.7' '10.3' '6.8' '19.3' '10.9' '17.1' '14.3'  
'11.1'  
'10.5' '9.1' '9.7' '9.8' '10.4' '9.2' '10.7' '14' '11.6' '12.3' '9.5']
```

```
'fe' '8.5' '12.8' '13.7' '11.5' '7.5' '7.6' '13']
```

Valeurs uniques pour 'salaire_net_mensuel_regional_1er_quartile':

```
['1620' '1460' '1450' '1370' '1430' 'nd' '1470' '1410' '1480' '1400'  
'1440' '1500' '1650' '1420' '.' '1490' '1380' '1640' '1520' '1510'  
'1350'
```

```
'1780' '1630' '1390' '1610' '1700' '1530' '1360' '1310' '1580' '1340'
```

nan

```
'1425' '1550' '1475' '1600' '1415' '1405' '1\xa0490']
```

Valeurs uniques pour 'salaire_net_mensuel_regional_3eme_quartile':

```
['2620' '2190' '2180' '2100' '2140' '2150' 'nd' '2330' '2240' '2220'  
'2120' '2160' '2170' '2270' '2210' '2700' '.' '2380' '2280' '2310'  
'2250'
```

```
'2260' '2360' '2790' '2480' '2300' '2320' '2290' '2390' '2340' '2060'
```

```
'2130' '2200' '2580' '2040' '2010' '2090' '2080' '2110' '2560' '2840'
```

```
'2510' '2020' '2030' '2530' '1980' '2070' '2230' '2050' nan '2420'
```

```
'2025'
```

```
'2570']
```

Valeurs uniques pour 'cle_etab':

```
['Paris 3 - Sorbonne Nouvelle_18' 'Paris 3 - Sorbonne Nouvelle_30'
```

```
'Sorbonne Université_30' 'Sorbonne Université_18'
```

```
'Université de Paris_30' 'Université de Paris_18' 'Rouen_18'
```

```
'Rouen_30'
```

```
'Le Havre_18' 'Le Havre_30' 'Paris-Est Marne-la-Vallée_30'
```

```
'Paris-Est Marne-la-Vallée_18' 'Versailles Saint-Quentin-en-
```

```
Yvelines_18'
```

```
'Versailles Saint-Quentin-en-Yvelines_30'
```

```
'Amiens - Picardie Jules-Verne_30' 'Amiens - Picardie Jules-Verne_18'
```

```
'Poitiers_18' 'Poitiers_30' 'Limoges_30' 'Limoges_18'
```

```
'Université de technologie de Belfort-Montbéliard_30' 'Paris-
```

```
Saclay_18'
```

```
'Paris-Saclay_30' 'Paris Nanterre_30' 'Paris Nanterre_18'
```

```
'Université Sorbonne Paris Nord_18' 'Université Sorbonne Paris
```

```
Nord_30'
```

```
'Paris 8 - Vincennes Saint-Denis_30' 'Paris 8 - Vincennes Saint-
```

```
Denis_18'
```

```
'Paris-Est Créteil_18' 'Paris-Est Créteil_30' 'Cergy-Pontoise_18'
```

```
'Cergy-Pontoise_30' 'Corse Pasquale Paoli_18' 'Corse Pasquale
```

```
Paoli_30'
```

```
'Antilles_18' 'Antilles_30' 'Guyane_18' 'Guyane_30' 'La Réunion_18'
```

```
'La Réunion_30' 'Polynésie Française_30'
```

```
'Toutes universités et établissements assimilés_30'
```

```
'Toutes universités et établissements assimilés_18'
```

```
'Nice - Sophia-Antipolis_30' 'Nice - Sophia-Antipolis_18'
```

```
'Université de technologie de Troyes_18' 'Aix-Marseille_30'
```

```
'Aix-Marseille_18' 'Caen Normandie_18' 'Caen Normandie_30'
```

```
'La Rochelle_30' 'Dijon - Bourgogne_30' 'Dijon - Bourgogne_18'
```

```
'Besançon - Franche-Comté_30' 'Besançon - Franche-Comté_18'
```


'COMUE UBFC_18' 'COMUE UBFC_30' 'Brest - Bretagne occidentale_30'
 'Brest - Bretagne occidentale_18' 'Nîmes_30' 'Nîmes_18'
 'Toulouse 1 - Capitole_30' 'Toulouse 3 - Paul Sabatier_18'
 'Toulouse 3 - Paul Sabatier_30' 'Bordeaux_18' 'Bordeaux_30'
 'Montpellier 3 - Paul-Valéry_18' 'Montpellier_18' 'Montpellier_30'
 'Rennes 1_18' 'Rennes 1_30' 'Rennes 2_30' 'Rennes 2_18'
 'Grenoble Alpes_30' 'Grenoble Alpes_18' 'Saint-Etienne - Jean
 Monnet_30'
 'Saint-Etienne - Jean Monnet_18' 'Orléans_18' 'Orléans_30'
 'Angers_30'
 'Angers_18' 'Reims - Champagne Ardenne_18' 'Reims - Champagne
 Ardenne_30'
 'Lorraine_30' 'Lorraine_18' 'Bretagne-Sud_18' 'Bretagne-Sud_30'
 'Université Polytechnique Hauts-de-France_18'
 'Université Polytechnique Hauts-de-France_30'
 "Littoral - Côte d'Opale_30" "Littoral - Côte d'Opale_18" 'Lille_30'
 'Lille_18' 'Artois_30' 'Artois_18' 'Clermont Auvergne_18'
 'Clermont Auvergne_30' "Pau - Pays de l'Adour_30"
 "Pau - Pays de l'Adour_18" 'Perpignan - Via Domitia_18'
 'Perpignan - Via Domitia_30' 'Strasbourg_30' 'Strasbourg_18'
 'Mulhouse - Haute Alsace_30' 'Mulhouse - Haute Alsace_18'
 'Lyon 1 - Claude Bernard_18' 'Lyon 1 - Claude Bernard_30'
 'Lyon 2 - Lumière_30' 'Lyon 2 - Lumière_18' 'Lyon 3 - Jean Moulin_30'
 'Lyon 3 - Jean Moulin_18' 'Le Mans_30' 'Le Mans_18'
 'Savoie Mont Blanc_18' 'Savoie Mont Blanc_30'
 'Paris 1 - Panthéon Sorbonne_18' 'Toulon_30' 'Toulon_18'
 "Évry-Val d'Essonne_18" 'Polynésie Française_18' "Côte d'Azur_30"
 "Côte d'Azur_18" 'Université de technologie de Troyes_30'
 'La Rochelle_18' 'Toulouse 1 - Capitole_18' 'Toulouse - Jean
 Jaures_30'
 'Bordeaux - Montaigne_30' 'Bordeaux - Montaigne_18'
 'Montpellier 3 - Paul-Valéry_30' 'Tours_18' 'Tours_30' 'Nantes_18'
 'Polytechnique Hauts-de-France_30' 'Polytechnique Hauts-de-France_18'
 'Paris 1 - Panthéon Sorbonne_30' 'Paris 2 - Panthéon Assas_30'
 'Paris 2 - Panthéon Assas_18' 'Gustave Eiffel_18' 'Gustave Eiffel_30'
 'INU Jean-François Champollion_18' 'INU Jean-François Champollion_30'
 'Avignon_30' "Évry-Val d'Essonne_30" 'CY Cergy Paris_30'
 'CY Cergy Paris_18' 'Toulouse - Jean Jaures_18' 'Nantes_30'
 'Paris 4 - Sorbonne_18' 'Paris René Descartes_30'
 'Paris 7 - Denis Diderot_30' 'Paris 7 - Denis Diderot_18'
 'Avignon_18'
 'Paris 11 - Paris Sud_30' 'Lille 1 - Sciences technologies_30'
 'Lille 1 - Sciences technologies_18' 'Lille 2 - Droit et santé_18'
 'Lille 2 - Droit et santé_30' 'Lille 3 - Charles-de-Gaulle_30'
 'Lille 3 - Charles-de-Gaulle_18' 'Paris 4 - Sorbonne_30'
 'Paris René Descartes_18' 'Paris Pierre et Marie Curie_18'
 'Paris Pierre et Marie Curie_30'
 'Université de technologie de Belfort-Montbéliard_18'
 'Paris 11 - Paris Sud_18' 'Clermont-Ferrand 2 - Blaise Pascal_18'

'Paris - Dauphine_18' 'Paris - Dauphine_30'
 'Clermont-Ferrand 1 - Auvergne_30' 'Clermont-Ferrand 1 - Auvergne_18'
 'Clermont-Ferrand 2 - Blaise Pascal_30'
 'Bordeaux 1 - Sciences technologies_30' 'Montpellier 1_30'
 'Montpellier 2 - Sciences techniques du Languedoc_30'
 'Grenoble 1 - Joseph Fourier_30' 'Grenoble 2 - Pierre Mendès
 France_30'
 'Grenoble 3 - Stendhal_30' 'Grenoble 1 - Joseph Fourier_18'
 'Grenoble 2 - Pierre Mendès France_18' 'Paris-Panthéon-Assas_30'
 'Grenoble 3 - Stendhal_18' 'Bordeaux 4 - Montesquieu_30'
 'Bordeaux 2 - Victor Segalen_30' 'Paris-Panthéon-Assas_18'
 'Lille Nord de France_18' 'Lille Nord de France_30']

Valeurs uniques pour 'cle_disc':

['disc06_18' 'disc06_30' 'disc18_30' 'disc01_30' 'disc16_18'
 'disc17_18'
 'disc12_30' 'disc13_18' 'disc13_30' 'disc18_18' 'disc03_18'
 'disc03_30'
 'disc04_30' 'disc08_30' 'disc15_18' 'disc02_30' 'disc14_30'
 'disc17_30'
 'disc01_18' 'disc09_18' 'disc09_30' 'disc04_18' 'disc05_30'
 'disc07_18'
 'disc11_18' 'disc10_30' 'disc12_18' 'disc14_18' 'disc15_30'
 'disc08_18'
 'disc07_30' 'disc02_18' 'disc11_30' 'disc16_30' 'disc10_18'
 'disc05_18'
 'disc19_30' 'disc20_30']

Valeurs uniques pour 'id_paysage':

['8k883' 'bxPQe' '5cZyU' 'G4572' 'LsQ24' '9Y8CD' 'V13Pk' '8j5s2'
 'hlX1r'
 'nkbwh' 'PVnB4' 'dV766' 'g6rwB' 'cqyN7' 'Uxr7Z' 'vb71K' 'W6CUQ'
 'NLCOF'
 'z3hdL' 'hy4EW' 'cEt92' 'zepT6' nan '7CYWd' '8tVLr' 'xJdyB' 'p25Q3'
 'atbEK' 'Lr940' '7Mpgt' '8A0mg' '06SE7' 'HAU8L' '542Id' 'm7K6T'
 '90I54'
 '1I7hJ' '7yd5d' '9wAap' 'ti37C' 'F2yh2' 'tIJ02' 'HCBvW' 'IXHyv'
 '570sX'
 't6Cq5' '5tVy4' 'SuUZ2' 'yH19Y' 'Szqy5' 'BWbvP' 'FvH4k' 'Mz286'
 'n1w55'
 '4k25D' '0JZ4a' 'etBz7' 'CUBKB' '7Gzub' '9xlel' 'zCa4j' '6G2TU'
 'C701f'
 'RN4E6' 's3t8T' 'HqAYu' 'Mz90U' 'evv7S' 'cqkij' 'Y7ch7' '7hB8r'
 'U8a0v'
 'EW53M' 'TWBzp' 'Hm42K' 'vxHYt' 'G2qA7' 'RS4WF' 'qUCQp' 'GBpkg'
 'kvciR'
 '02v0H' '5KpJ9' 'X58LX' 'u13se' 'XDl91' 'qPCgk' 'j5sy1' 'QhvwL'
 '3RMeh'
 'K4lR3' 'n1E6T' 'C6Ps7' '3itX6' 'IXJPr' '9ZF2M' 'w3F5n' 'Mrr0h'

```
'VrYMi'  
'Xfv3j' 'vFsty' 'xE7KR' 'IZ3tY']
```

```
print(df.isna().sum())
```

```
annee                                0  
diplome                             0  
numero_de_l_etablissement           0  
etablissement                       0  
etablissementactuel                 15149  
code_de_l_academie                  361  
academie                           361  
code_du_domaine                     0  
domaine                             0  
code_de_la_discipline                0  
discipline                          0  
situation                           0  
remarque                            18034  
nombre_de_reponses                  47  
taux_de_reponse                      321  
poids_de_la_discipline               838  
taux_dinsertion                      8747  
taux_d_emploi                       17669  
taux_d_emploi_salarie_en_france     17669  
emplois_cadre_ou_professions_intermediaires 9787  
emplois_stables                     9668  
emplois_a_temps_plein               9734  
salaire_net_median_des_emplois_a_temps_plein 11341  
salaire_brut_annuel_estime           11161  
de_diplomes_boursiers               198  
taux_de_chomage_regional             0  
salaire_net_mensuel_median_regional  682  
emplois_cadre                       10262  
emplois_exterieurs_a_la_region_de_luniversite 10067  
femmes                              9245  
salaire_net_mensuel_regional_1er_quartile 2025  
salaire_net_mensuel_regional_3eme_quartile 2025  
cle_etab                             0  
cle_disc                             0  
id_paysage                           361  
dtype: int64
```

```
drop_duplicates = df.drop_duplicates()
```

```
df.drop(  
    columns=[  
        'etablissementactuel',  
        'remarque',  
        'taux_d_emploi',
```

```

        'taux_d_emploi_salarie_en_france',
        'cle_etab',
        'cle_disc',
        'id_paysage',
        'salaire_net_mensuel_regional_1er_quartile',
        'salaire_net_mensuel_regional_3eme_quartile',
        'taux_de_reponse',
        'nombre_de_reponses',
        'code_du_domaine',
        'taux_de_chomage_regional',
        'code_de_la_discipline',
        'code_de_l_academie',
        'numero_de_l_etablissement'
    ],
    inplace=True
)

df.shape
(19603, 19)

df.columns
Index(['annee', 'diplome', 'etablissement', 'academie', 'domaine',
       'discipline', 'situation', 'poids_de_la_discipline',
       'taux_dinsertion',
       'emplois_cadre_ou_professions_intermediaires',
       'emplois_stables',
       'emplois_a_temps_plein',
       'salaire_net_median_des_emplois_a_temps_plein',
       'salaire_brut_annuel_estime', 'de_diplomes_boursiers',
       'salaire_net_mensuel_median_regional', 'emplois_cadre',
       'emplois_exterieurs_a_la_region_de_luniversite', 'femmes'],
      dtype='object')

print(df.isna().sum())

```

annee	0
diplome	0
etablissement	0
academie	361
domaine	0
discipline	0
situation	0
poids_de_la_discipline	838
taux_dinsertion	8747
emplois_cadre_ou_professions_intermediaires	9787
emplois_stables	9668
emplois_a_temps_plein	9734
salaire_net_median_des_emplois_a_temps_plein	11341
salaire_brut_annuel_estime	11161

de_diplomes_boursiers	198
salaire_net_mensuel_median_regional	682
emplois_cadre	10262
emplois_exterieurs_a_la_region_de_luniversite	10067
femmes	9245
dtype: int64	

```
df = df[~df.isin(['ns', 'nd']).any(axis=1)]
```

ns = donnée non significative

nd = donnée non dispo

```
df.shape
```

```
(19603, 19)
```

```
print(df.isna().sum())
```

annee	0
diplome	0
etablissement	0
academie	361
domaine	0
discipline	0
situation	0
poids_de_la_discipline	838
taux_dinsertion	8747
emplois_cadre_ou_professions_intermediaires	9787
emplois_stables	9668
emplois_a_temps_plein	9734
salaire_net_median_des_emplois_a_temps_plein	11341
salaire_brut_annuel_estime	11161
de_diplomes_boursiers	198
salaire_net_mensuel_median_regional	682
emplois_cadre	10262
emplois_exterieurs_a_la_region_de_luniversite	10067
femmes	9245
dtype: int64	

```
print(df.dtypes)
```

annee	int64
diplome	object
etablissement	object
academie	object
domaine	object
discipline	object
situation	object
poids_de_la_discipline	float64
taux_dinsertion	float64

```

emplois_cadre_ou_professions_intermediaires    float64
emplois_stables                                float64
emplois_a_temps_plein                          float64
salaire_net_median_des_emplois_a_temps_plein   float64
salaire_brut_annuel_estime                     float64
de_diplomes_boursiers                         float64
salaire_net_mensuel_median_regional            float64
emplois_cadre                                  float64
emplois_exterieurs_a_la_region_de_luniversite  float64
femmes                                          float64
dtype: object

```

```

for col in df.select_dtypes(include='float64').columns:
    unique_values = df[col].unique()
    print(f"Valeurs uniques pour '{col}':\n{unique_values}\n")

```

Valeurs uniques pour 'poids_de_la_discipline':

```

[ 57. 100.  1.  11.  10.  15.  33.  13.  6.  3.  24.  5.  12.  67.
  22.  2.  0.  4.  72.  39.  36.  16.  7.  28. nan  31.  8.  61.
  26.  30.  42.  21.  20.  9.  18.  14.  47.  23.  45.  35.  37.  55.
  34.  17.  19.  25.  32.  90.  60.  40.  62.  46.  64.  59.  29.  27.
  38.  44.  73.  56.  58.  63.  50.  43.  53.  65.  41.  49.  91.  70.
  52.  48.  89.  88.  66.  54.  83.  71.  76.  77.  68.  51.  87.  86.
  92.  84.  78.  80.  96.  69.  85.  74.  93.  82.  81.  97.  75.]

```

Valeurs uniques pour 'taux_dinsertion':

```

[ 92.  91.  99. nan  96.  97.  98.  90. 100.  93.  84.  85.  88.  94.
  95.  77.  78.  83.  82.  87.  89.  86.  79.  81.  73.  74.  67.  80.
  72.  71.  69.  76.  68.  64.  75.  66.  70.  62.  63.  65.  34.  37.
  60.  13.  6.  61.  56.]

```

Valeurs uniques pour 'emplois_cadre_ou_professions_intermediaires':

```

[ 72.  80.  98. nan  93. 100.  97.  99.  91.  94.  95.  84.  96.  76.
  85.  83.  89.  86.  74.  92.  81.  87.  90.  63.  88.  82.  79.  52.
  69.  67.  75.  70.  60.  59.  64.  73.  78.  65.  62.  66.  55.  77.
  57.  71.  68.  61.  45.  49.  58.  53.  56.  51.  11.  26.  21.  18.
  44.  43.  54.  48.  37.  39.  50.  3.  36.  47.  32.  41.  46.
  20.]

```

Valeurs uniques pour 'emplois_stables':

```

[ 58.  66.  97. nan  89.  79.  81.  52.  70. 100.  87.  76.  32.  88.
  64.  65.  85.  93.  54.  59.  53.  57.  86.  44.  67.  61.  83.  77.
  69.  68.  74.  73.  82.  90.  71.  78.  94.  96.  84.  92.  51.  75.
  46.  72.  45.  91.  63.  38.  80.  50.  55.  49.  36.  42.  60.  62.
  98.  27.  95.  56.  48.  99.  35.  23.  47.  37.  31.  26.  39.  33.
  25.  41.  28.  40.  29.  43.  34.  24.  15.  30.  22.  21.  19.  18.
  12.]

```

Valeurs uniques pour 'emplois_a_temps_plein':

```

[ 83.  87.  95. nan  99. 100.  98.  97.  96.  93.  90.  64.  79.  74.]

```

91.	89.	77.	80.	94.	82.	81.	75.	63.	84.	85.	73.	92.	88.
72.	62.	76.	78.	67.	70.	69.	86.	60.	45.	68.	39.	61.	51.
71.	48.	56.	65.	58.	42.	53.	59.	57.	55.	66.	54.	44.	52.
37.	47.	50.	46.	41.]									

Valeurs uniques pour 'salaire_net_median_des_emplois_a_temps_plein':

[1500. 1800. 1820. nan 2400. 2340. 2440. 2120. 2390. 2370. 1700.
2080.
1790. 1900. 2100. 2350. 1750. 1880. 2070. 1980. 1600. 1720. 2000.
2180.
1870. 1890. 2250. 2200. 1740. 2110. 2500. 2600. 2700. 2140. 2150.
1950.
1690. 2330. 2040. 1650. 1860. 1970. 1580. 2220. 1940. 1960. 1630.
2010.
1930. 2050. 1840. 1680. 1520. 2170. 1730. 2310. 2280. 2030. 2270.
2060.
1850. 1830. 1620. 1760. 1610. 1990. 2300. 1640. 2160. 2090. 2290.
2190.
2210. 1780. 1920. 2230. 2020. 2240. 1560. 1670. 1530. 2430. 2130.
1660.
2680. 1710. 2380. 1770. 1570. 1550. 2530. 2560. 2690. 2650. 2800.
2360.
2770. 2990. 1810. 2490. 1590. 2450. 1540. 1400. 1490. 1470. 1480.
2320.
2750. 2810. 2470. 2480. 1510. 1910. 1410. 1420. 1450. 1380. 1460.
1350.
2580. 2540. 2410. 2420. 2520. 2900. 2550. 2640. 1280. 1440. 2260.
1430.
1300. 2620. 2710. 2460. 3100. 2670. 3000. 2760. 2980. 2510. 1330.
2720.
3150. 2630. 2570. 1370. 2740. 2610. 2730. 1360. 2920. 2930. 2590.
2830.]

Valeurs uniques pour 'salaire_brut_annuel_estime':

[23400. 28100. 28300. nan 37400. 36500. 38100. 33100. 37300. 37000.
26500. 33000. 32500. 27900. 29600. 32800. 36700. 27300. 29300. 32300.
30900. 24900. 26800. 31200. 34000. 29100. 29500. 35100. 34300. 27100.
39000. 40600. 42100. 33400. 33500. 30400. 26400. 36300. 33300. 31800.
28000. 25700. 29000. 30700. 24600. 34700. 30200. 30500. 25400. 31300.
30100. 32000. 28700. 26200. 23700. 33800. 30800. 26900. 36100. 32200.
35600. 31700. 35500. 35400. 36000. 32100. 32900. 28800. 28600. 25300.
27500. 29400. 25200. 31000. 35900. 25600. 33700. 28900. 32700. 32600.
33900. 35700. 34100. 34500. 27800. 29900. 34800. 31400. 26300. 25000.
27200. 34900. 24300. 28500. 26000. 27000. 23900. 37900. 25500. 30000.
31500. 33200. 29200. 25900. 35800. 41800. 27400. 26600. 35300. 34600.
37200. 26700. 27700. 24400. 24200. 32400. 39500. 39900. 42000. 41300.
35200. 26100. 31600. 43700. 25100. 36800. 30600. 35000. 43200. 46600.
28200. 38900. 24700. 27600. 38200. 36400. 24100. 24500. 21800. 23200.
28400. 22900. 23800. 23000. 31100. 36200. 24800. 38800. 30300. 42900.]

```

43800. 38500. 38700. 34200. 23500. 29800. 22000. 23100. 22100. 37100.
22600. 25800. 31900. 22200. 21500. 22800. 23600. 29700. 21100. 33600.
40200. 22700. 24000. 39700. 37600. 34400. 23300. 37700. 36900. 39300.
45200. 39800. 41200. 20000. 22500. 38600. 22300. 22400. 20300. 40800.
42200. 38400. 48400. 41900. 38000. 41600. 46800. 43100. 46500. 37800.
40300. 39200. 20800. 20700. 21900. 39600. 42400. 49100. 41000. 38300.
39100. 40000. 21300. 21600. 42700. 40700. 42600. 21200. 41700. 36600.
39400. 45500. 45600. 40500. 41100. 40400. 44200.]

```

Valeurs uniques pour 'de_diplomes_boursiers':

```

[ 33.  16.  26.  27.  14.  36.  17.  21.  32.  35.
 34.  50.  18.  28.  38.  29.  31.    nan  49.   9.
 11.  30.  39.  41.  20.  63.   0.  24.  53.  48.
 40.  44.   5.  22.  23.  37.  43.   8.  15.  42.
 13.  54. 100.  51.  19.  60.  56.  46.  25.  12.
 10.  12.93 23.76 22.89 35.36 30.08 32.7 43.72 26.4 36.11
 37.41 48.09 26.47 49.25 30.81 45.72 32.49 50.32 30.34 29.98
 28.42 49.55 26.52 32.56 29.13 33.9 23.08 29.29 39.82 21.16
 43.46 34.5 35.4 21.24 21.21 29.4 45.   7.  47.  58.
 64.  52.  27.1 27.63 31.25 35.89 42.34 6.67 41.65 27.46
 21.29 27.69 38.42 23.78 43.5 14.22 28.17 21.45 29.5 27.19
 20.78 32.08 27.78 36.8  9.89 29.81 57.68 40.09 43.53 30.55
 46.75 23.16 11.41 16.36 28.61 34.85 30.78 8.23 32.31 7.57
 16.4 26.78 41.92 36.05 48.87 37.3 26.92 21.67 30.99 62.
 67.   6.  66.   2.   4.  31.52 27.82 29.57 27.39 32.77
 22.88 13.43 26.91 7.41 23.81 2.59 61.  57.  65.   1.
 25.44 24.22 24.54 6.45 35.71 28.57 3.  10.34 8.82 7.69
 22.22 18.84 29.99 14.08 17.24 18.07 7.98 33.05 26.15 20.18
 40.53 29.03 24.98 29.21 15.63 21.82]

```

Valeurs uniques pour 'salaire_net_mensuel_median_regional':

```

[2120. 1840. 1850. 1760. 1810. 1830.    nan 1900. 1860. 1780. 1820.
 1790.
 1890. 1870. 2170. 1930. 1880. 1910. 1920. 2240. 1980. 1940. 1950.
 1730.
 2070. 1800. 1750. 2050. 2300. 2020. 1740. 1710. 1680. 1720. 1670.
 1650.
 1690. 1990. 1660. 2090. 1700. 1770. 1745. 1725. 1825. 2060. 1865.
 1580.]

```

Valeurs uniques pour 'emplois_cadre':

```

[ 46.  57.  96.  nan  93.  98.  89.  91.  90.  97.  59.  75.  70.  52.
 81.  40.  87.  34.  74.  84.  67.  61.  82.  83.  51. 100.  78.  55.
 79.  65.  56.  69.  32.  62.  68.  60.  94.  66.  80.  58.  64.  71.
 54.  26.  53.  45.  38.  63.  77.  31.  49.  72.  42.  50.  95.  73.
 48.  27.  76.  44.  41.  35.  29.  33.  92.  39.  86.  37.  88.  36.
 20.  43.  47.  85.  28.  22.  19.   2.  10.  18.   1.   6.  30.  16.
 24.  99.  15.  23.  25.  17.  21.  13.  11.   5.  12.   3.   9.  14.
 7.]

```



```
Valeurs uniques pour 'emplois_exterieurs_a_la_region_de_luniversite':
[ 35.   6.  nan  15.   8.  10.  16.  26.   5.  12.  37.  44.  22.  48.
  11.  30.  19.  63.  47.  79.  62.  83.  51.  59.  69.  14.  27.  38.
  41.  17.  23.  36. 100.  24.  42.  43.  39.  32.  52.  33.  25.  50.
  54.  78.  56.  21.  46.  61.  53.  45.  65.  73.  58.  49.  64.  55.
  60.  20.  40.  75.  57.  68.  31.  29.   9.   3.  28.  13.  70.  72.
  34.  66.  71.  76.  67.  91.  18.  82.   4.  80.   7.  84.   2.  74.
  81.  98.  97.  87.  93.  77.  88.  85.  96.  86.  90.  95.  92.]
```

```
Valeurs uniques pour 'femmes':
[ 89.  86.  nan   9.  32.  58.  46.  68.  10.  72.  61.  36.  62.  52.
  74.  66.  82.  73.  43.  28.  42.  67.  65.  44.  69.  45.  83.  84.
  54.  40.  71.  59.  51.  18.  77.  63.  76.  60.  13.  75.  53.  15.
  55.  94.  64.  79.  93.   3.  29.  57.  87.  38.  11.  16.  47.  56.
  25.  22.  31.  27.  39.  81.  34.  33.  21.  35.  78.  70.  80.  88.
  37.  90.  48.  30.  50.  41.  12.   5.  97.  85.  92.  20.  49.  17.
  91.   2.   7.  23.   4.   8.  95.   6.  26.  14.  24.  19.  96.  98.
 100.]
```

```
df = df.dropna()
```

```
df.shape
```

```
(7346, 19)
```

```
df.isna().sum()
```

```
annee                                0
diplome                              0
etablissement                        0
academie                             0
domaine                             0
discipline                           0
situation                            0
poids_de_la_discipline               0
taux_dinsertion                      0
emplois_cadre_ou_professions_intermediaires  0
emplois_stables                      0
emplois_a_temps_plein                0
salaire_net_median_des_emplois_a_temps_plein  0
salaire_brut_annuel_estime            0
de_diplomes_boursiers                0
salaire_net_mensuel_median_regional    0
emplois_cadre                        0
emplois_exterieurs_a_la_region_de_luniversite  0
femmes                               0
dtype: int64
```

```
df
```

```

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69.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\":\n          \"femmes\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 19.176047452353345,\n            \"min\": 2.0,\n            \"max\": 98.0,\n            \"num_unique_values\": 96,\n            \"samples\": [\n              23.0,\n              41.0\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          }\n        }\n      ],\n      \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```
df[df['annee'] >= 2017]
```

```

{"summary": "{\n  \"name\": \"df[df['annee'] >= 2017]\",\n  \"rows\": 3081,\n  \"fields\": [\n    {\n      \"column\": \"annee\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 1,\n        \"min\": 2017,\n        \"max\": 2020,\n        \"num_unique_values\": 4,\n        \"samples\": [\n          2018,\n          2020,\n          2017\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\": \"diplome\",\n        \"properties\": {\n          \"dtype\": \"category\",\n          \"num_unique_values\": 2,\n          \"samples\": [\n            \"MASTER ENS\",\n            \"MASTER LMD\"\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"etablissement\",\n          \"properties\": {\n            \"dtype\": \"category\",\n            \"num_unique_values\": 67,\n            \"samples\": [\n              \"Universit\u00e9 Polytechnique Hauts-de-France\",\n              \"Nice - Sophia-Antipolis\"\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          },\n          {\n            \"column\": \"academie\",\n            \"properties\": {\n              \"dtype\": \"category\",\n              \"num_unique_values\": 25,\n              \"samples\": [\n                \"Aix-Marseille\",\n                \"Lyon\"\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n              \"column\": \"domaine\",\n              \"properties\": {\n                \"dtype\": \"category\",\n                \"num_unique_values\": 5,\n                \"samples\": [\n                  \"Masters enseignement\",\n                  \"Sciences humaines et sociales\"\n                ],\n                \"semantic_type\": \"\",\n                \"description\": \"\"\n              },\n              {\n                \"column\": \"discipline\",\n                \"properties\": {\n                  \"dtype\": \"category\",\n                  \"num_unique_values\": 18,\n                  \"samples\": [\n                    \"Masters enseignement\"\n                  ],\n                  \"semantic_type\": \"\",\n                  \"description\": \"\"\n                },\n                {\n                  \"column\": \"situation\",\n                  \"properties\": {\n                    \"dtype\": \"category\",\n                    \"num_unique_values\": 2,\n                    \"samples\": [\n                      \"30 mois apr\u00e8s le dipl\u00f4me\",\n                      \"18 mois apr\u00e8s le dipl\u00f4me\"\n                    ],\n                    \"semantic_type\": \"\",\n                    \"description\": \"\"\n                  },\n                  {\n                    \"column\": \"poids_de_la_discipline\",\n                    \"properties\": {\n                      \"dtype\": \"number\",\n                      \"std\":

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26.603595244697264,\n          \"min\": 2.0,\n          \"max\": 100.0,\n          \"num_unique_values\": 83,\n          \"samples\": [\n          17.0,\n          57.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"taux_dinsertion\",\n          \"properties\": {\n          \"dtype\":\n          \"number\",\n          \"std\": 5.299051663503572,\n          \"min\":\n          67.0,\n          \"max\": 100.0,\n          \"num_unique_values\": 33,\n          \"samples\": [\n          76.0,\n          82.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          }\n          },\n          {\n          \"column\":\n          \"emplois_cadre_ou_professions_intermediaires\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\":\n          8.292683825135757,\n          \"min\": 46.0,\n          \"max\": 100.0,\n          \"num_unique_values\": 50,\n          \"samples\": [\n          71.0,\n          85.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"emplois_stables\",\n          \"properties\": {\n          \"dtype\":\n          \"number\",\n          \"std\": 14.740889736390173,\n          \"min\":\n          19.0,\n          \"max\": 100.0,\n          \"num_unique_values\": 73,\n          \"samples\": [\n          79.0,\n          37.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          }\n          },\n          {\n          \"column\": \"emplois_a_temps_plein\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\":\n          6.920511552518593,\n          \"min\": 57.0,\n          \"max\": 100.0,\n          \"num_unique_values\": 42,\n          \"samples\": [\n          82.0,\n          89.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"salaire_net_median_des_emplois_a_temps_plein\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\":\n          262.80579205810443,\n          \"min\": 1360.0,\n          \"max\":\n          3150.0,\n          \"num_unique_values\": 140,\n          \"samples\": [\n          2320.0,\n          2020.0\n          ],\n          \"semantic_type\":\n          \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"salaire_brut_annuel_estime\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\":\n          4100.255336683942,\n          \"min\": 21200.0,\n          \"max\": 49100.0,\n          \"num_unique_values\": 203,\n          \"samples\": [\n          36700.0,\n          26500.0\n          ],\n          \"semantic_type\":\n          \"\",\n          \"description\": \"\"\n          },\n          {\n          \"column\":\n          \"de_diplomes_boursiers\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\":\n          8.700151435592051,\n          \"min\": 0.0,\n          \"max\": 56.0,\n          \"num_unique_values\":\n          137,\n          \"samples\": [\n          27.82,\n          43.72\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n          }\n          },\n          {\n          \"column\":\n          \"salaire_net_mensuel_median_regional\",\n          \"properties\": {\n          \"dtype\": \"number\",\n          \"std\":\n          145.5297660863258,\n          \"min\": 1760.0,\n          \"max\": 2300.0,\n          \"num_unique_values\": 25,\n          \"samples\": [\n          1810.0,\n

```

```

n          1900.0\n          ],\n          \"semantic_type\": \"\", \n\n\"description\": \"\"\n          },\n          {\n          \"column\":\n\n\"emplois_cadre\", \n          \"properties\": {\n          \"dtype\":\n\n\"number\", \n          \"std\": 18.100302243194584, \n          \"min\":\n\n1.0, \n          \"max\": 100.0, \n          \"num_unique_values\": 91, \n          \"samples\": [\n          45.0, \n          74.0\n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\"\n          }\n          },\n          {\n          \"column\":\n\n\"emplois_exterieurs_a_la_region_de_luniversite\", \n          \"properties\": {\n          \"dtype\": \"number\", \n          \"std\":\n\n17.183992546891567, \n          \"min\": 2.0, \n          \"max\": 93.0, \n          \"num_unique_values\": 86, \n          \"samples\": [\n          80.0, \n          35.0\n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\"\n          }, \n          {\n          \"column\":\n\n\"femmes\", \n          \"properties\": {\n          \"dtype\": \"number\", \n          \"std\":\n\n19.122124296514098, \n          \"min\": 2.0, \n          \"max\":\n\n97.0, \n          \"num_unique_values\": 94, \n          \"samples\": [\n          79.0, \n          69.0\n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\"\n          }\n          }\n          ]\n          }\", \"type\": \"dataframe\"}

```

```
round(df.isna().sum() / len(df) * 100)
```

annee	0.0
diplome	0.0
etablissement	0.0
academie	0.0
domaine	0.0
discipline	0.0
situation	0.0
poids_de_la_discipline	0.0
taux_dinsertion	0.0
emplois_cadre_ou_professions_intermediaires	0.0
emplois_stables	0.0
emplois_a_temps_plein	0.0
salaire_net_median_des_emplois_a_temps_plein	0.0
salaire_brut_annuel_estime	0.0
de_diplomes_boursiers	0.0
salaire_net_mensuel_median_regional	0.0
emplois_cadre	0.0
emplois_exterieurs_a_la_region_de_luniversite	0.0
femmes	0.0
dtype: float64	

Calculations

```
df.describe()
```

```

{\"summary\": \"{\\n  \"name\": \"df\", \\n  \"rows\": 8, \\n  \"fields\": [\\n
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\\n      \"dtype\": \"number\", \\n      \"std\": 2107.4370689434018, \\n

```

```

{"min": 2.725999792113035,\n
 "num_unique_values": 8,\n
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  2016.0,\n
  7346.0\n
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 "semantic_type": "\"",\n
 "description": "\""\n
 },\n
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 "properties": {\n
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 "std": 2585.7649600708937,\n
 "min": 2.0,\n
 "max": 7346.0,\n
 "num_unique_values": 8,\n
 "samples": [\n
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  22.0,\n
  7346.0\n
 ],\n
 "semantic_type": "\"",\n
 "description": "\""\n
 },\n
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 "properties": {\n
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 "min": 5.806050159948711,\n
 "max": 7346.0,\n
 "num_unique_values": 8,\n
 "samples": [\n
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  91.0,\n
  7346.0\n
 ],\n
 "semantic_type": "\"",\n
 "description": "\""\n
 },\n
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 "max": 7346.0,\n
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  91.0,\n
  7346.0\n
 ],\n
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 "description": "\""\n
 },\n
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 "std": 2576.008276893464,\n
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 "max": 7346.0,\n
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 "samples": [\n
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  73.0,\n
  7346.0\n
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 "description": "\""\n
 },\n
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 "max": 7346.0,\n
 "num_unique_values": 8,\n
 "samples": [\n
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  96.0,\n
  7346.0\n
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 "description": "\""\n
 },\n
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 "num_unique_values": 8,\n
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  1900.0,\n
  7346.0\n
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 "description": "\""\n
 },\n
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 "min": 4003.3250718655077,\n
 "max": 49100.0,\n
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 "samples": [\n
  29915.8589708685,\n
  29600.0,\n
  7346.0\n
 ],\n
 "semantic_type": "\"",\n
 "description": "\""}

```



```

{"semantic_type": "\",\n      \"description\": \"\"\n    },\n    {\n      \"column\": \"de_diplomes_boursiers\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 2587.4505544043077,\n        \"min\": 0.0,\n        \"max\": 7346.0,\n        \"num_unique_values\": 8,\n        \"samples\": [\n          30.19713721753335,\n          30.34,\n          7346.0\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\": \"salaire_net_mensuel_median_regional\",\n        \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 2119.789253879192,\n          \"min\": 136.30629006623818,\n          \"max\": 7346.0,\n          \"num_unique_values\": 8,\n          \"samples\": [\n            1873.48897359107,\n            1830.0,\n            7346.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"emplois_cadre\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 2578.2435817812006,\n            \"min\": 1.0,\n            \"max\": 7346.0,\n            \"num_unique_values\": 8,\n            \"samples\": [\n              64.6722025592159,\n              65.0,\n              7346.0\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          },\n          {\n            \"column\": \"emplois_exterieurs_a_la_region_de_luniversite\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 2582.8354039419305,\n              \"min\": 2.0,\n              \"max\": 7346.0,\n              \"num_unique_values\": 8,\n              \"samples\": [\n                41.75646610400218,\n                43.0,\n                7346.0\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n              \"column\": \"femmes\",\n              \"properties\": {\n                \"dtype\": \"number\",\n                \"std\": 2579.2762164886344,\n                \"min\": 2.0,\n                \"max\": 7346.0,\n                \"num_unique_values\": 8,\n                \"samples\": [\n                  58.18622379526273,\n                  60.0,\n                  7346.0\n                ],\n                \"semantic_type\": \"\",\n                \"description\": \"\"\n              }\n            }\n          }\n        }\n      ],\n      \"type\": \"dataframe\"}

```

```

discipline_groupby_1 = df.groupby('discipline')
['taux_dinsertion'].mean()
discipline_groupby_sorted =
discipline_groupby_1.sort_values(ascending=True)
print(discipline_groupby_sorted)

```

discipline	
Histoire-géographie	84.655556
Lettres, langues, arts	85.487871
Sciences de la vie et de la terre	85.739474
Autres sciences humaines et sociales	86.434483
Ensemble sciences humaines et sociales	86.858711
Information communication	88.155340
Sciences fondamentales	89.418269
Autres formations juridiques, économiques et de gestion	90.006135

Économie	90.028455
Ensemble sciences, technologies et santé	90.456140
Ensemble formations juridiques, économiques et de gestion	91.335031
Psychologie	91.673469
Autres sciences, technologies et santé	91.808411
Droit	91.971429
Sciences de l'ingénieur	92.360714
Gestion	92.802781
Informatique	95.974265
Masters enseignement : premier degré	96.214286
Masters enseignement : second degré, CPE...	97.307692
Masters enseignement	97.643617

Name: taux_dinsertion, dtype: float64

```
discipline_groupby_2 = df.groupby('discipline')
['salaire_brut_annuel_estime'].mean()
discipline_groupby_sorted_2 =
discipline_groupby_2.sort_values(ascending=True)
print(discipline_groupby_sorted_2)
```

```
discipline
Masters enseignement : premier degré
25771.428571
Lettres, langues, arts
25842.857143
Ensemble sciences humaines et sociales
26593.689986
Psychologie
27071.428571
Autres sciences humaines et sociales
27138.275862
Information communication
27352.427184
Masters enseignement
27569.326241
Histoire-géographie
28062.222222
Masters enseignement : second degré, CPE...
28126.923077
Sciences de la vie et de la terre
28492.105263
Autres sciences, technologies et santé
29738.317757
Droit
30427.472527
Ensemble formations juridiques, économiques et de gestion
31024.134420
Autres formations juridiques, économiques et de gestion
31234.969325
Ensemble sciences, technologies et santé
```

```

31423.742690
Gestion
32107.711757
Économie
32831.707317
Sciences fondamentales
33037.980769
Sciences de l'ingénieur
33042.857143
Informatique
34527.941176
Name: salaire_brut_annuel_estime, dtype: float64

groupby_1 = df.groupby('discipline')['emplois_stables'].mean()
groupby_sorted3 = groupby_1.sort_values(ascending=True)

print("Average stable employment by")
print(groupby_sorted3)

Average stable employment by
discipline
Autres sciences humaines et sociales      52.989655
Histoire-géographie                        53.288889
Ensemble sciences humaines et sociales    54.534979
Sciences de la vie et de la terre         57.363158
Psychologie                              59.040816
Information communication                  61.975728
Lettres, langues, arts                     62.053908
Autres formations juridiques, économiques et de gestion 67.564417
Droit                                     70.147253
Autres sciences, technologies et santé    72.785047
Économie                                 74.341463
Ensemble formations juridiques, économiques et de gestion 75.070265
Masters enseignement : premier degré      75.178571
Sciences fondamentales                    75.192308
Ensemble sciences, technologies et santé  75.419883
Gestion                                  80.251580
Sciences de l'ingénieur                    84.857143
Masters enseignement                      86.590426
Masters enseignement : second degré, CPE... 88.576923
Informatique                             92.591912
Name: emplois_stables, dtype: float64

```

Regression simple

```

univariate_regression = smf.ols(formula='taux_dinsertion ~
discipline', data=df).fit()
print(univariate_regression.summary())

```

OLS Regression Results

```

=====
Dep. Variable:          taux_dinsertion    R-squared:
0.336
Model:                  OLS               Adj. R-squared:
0.335
Method:                 Least Squares     F-statistic:
195.3
Date:                   Mon, 18 Nov 2024   Prob (F-statistic):
0.00
Time:                   16:17:58          Log-Likelihood:
-21839.
No. Observations:      7346              AIC:
4.372e+04
Df Residuals:          7326              BIC:
4.386e+04
Df Model:               19
Covariance Type:       nonrobust

```

```

=====
=====
coef      std err      t      P>|t|      [0.025      0.975]
-----
Intercept
90.0061    0.371    242.615    0.000    89.279    90.733
discipline[T.Autres sciences humaines et sociales]
-3.5717    0.464    -7.703    0.000    -4.481    -2.663
discipline[T.Autres sciences, technologies et santé]
1.8023     0.492     3.660    0.000     0.837     2.768
discipline[T.Droit]
1.9653     0.432     4.546    0.000     1.118     2.813
discipline[T.Ensemble formations juridiques, économiques et de
gestion]    1.3289     0.401     3.317    0.001     0.544     2.114
discipline[T.Ensemble sciences humaines et sociales]
-3.1474    0.410    -7.670    0.000    -3.952    -2.343
discipline[T.Ensemble sciences, technologies et santé]
0.4500     0.405     1.112    0.266    -0.344     1.244
discipline[T.Gestion]
2.7966     0.407     6.864    0.000     1.998     3.595
discipline[T.Histoire-géographie]
-5.3506    0.622    -8.602    0.000    -6.570    -4.131
discipline[T.Information communication]
-1.8508    0.497    -3.728    0.000    -2.824    -0.877
discipline[T.Informatique]

```

5.9681	0.469	12.721	0.000	5.048	6.888
discipline[T.Lettres, langues, arts]					
-4.5183	0.445	-10.152	0.000	-5.391	-3.646
discipline[T.Masters enseignement]					
7.6375	0.421	18.133	0.000	6.812	8.463
discipline[T.Masters enseignement : premier degré]					
6.2082	0.969	6.407	0.000	4.309	8.108
discipline[T.Masters enseignement : second degré, CPE...]					
7.3016	1.000	7.300	0.000	5.341	9.262
discipline[T.Psychologie]					
1.6673	0.502	3.321	0.001	0.683	2.652
discipline[T.Sciences de l'ingénieur]					
2.3546	0.467	5.046	0.000	1.440	3.269
discipline[T.Sciences de la vie et de la terre]					
-4.2667	0.443	-9.621	0.000	-5.136	-3.397
discipline[T.Sciences fondamentales]					
-0.5879	0.495	-1.187	0.235	-1.559	0.383
discipline[T.Économie]					
0.0223	0.478	0.047	0.963	-0.915	0.960

```

=====
=====
Omnibus:                649.342    Durbin-Watson:
1.684
Prob(Omnibus):          0.000    Jarque-Bera (JB):
991.576
Skew:                   -0.679    Prob(JB):
4.81e-216
Kurtosis:               4.180    Cond. No.
32.1
=====
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Interprétation des résultats de la régression OLS

R-squared (R^2) : 0.336

Interprétation : Cela signifie que 33,6 % de la variance du taux d'insertion (taux_dinsertion) est expliquée par le modèle de régression. Un R^2 de 0.336 indique un modèle modérément performant, mais il reste 66,4 % de la variance inexpliquée, ce qui suggère qu'il existe d'autres facteurs non inclus dans le modèle qui influencent le taux d'insertion.

F-statistic : 195.3, Prob (F-statistic) : 0.00

Interprétation : La valeur F élevée (195.3) et la probabilité de 0,00 indiquent que le modèle est statistiquement significatif, c'est-à-dire que les variables explicatives (ici, les disciplines) ont un effet global significatif sur le taux d'insertion.

Intercept (constante) : 90.0061

Interprétation : Lorsque toutes les disciplines sont égales à zéro (c'est-à-dire la catégorie de référence), le taux d'insertion estimé est de 90.0061 %. En d'autres termes, le taux d'insertion pour la catégorie de discipline de référence (celle exclue des variables indicatrices) est de 90,01 %.

Coefficients des différentes disciplines

Les coefficients indiquent l'impact des différentes disciplines sur le taux d'insertion par rapport à la discipline de référence. Voici une interprétation de quelques disciplines :

Autres sciences humaines et sociales : -3.5717

Cela signifie que, par rapport à la discipline de référence, le taux d'insertion des diplômés dans cette discipline est inférieur de 3.5717 %.

La valeur p associée (0.000) montre que cet effet est statistiquement significatif.

Autres sciences, technologies et santé : 1.8023

Le taux d'insertion des diplômés dans cette discipline est supérieur de 1.8023 % par rapport à la discipline de référence.

La valeur p associée (0.000) montre également que cet effet est statistiquement significatif.

Droit : 1.9653

Les diplômés en Droit ont un taux d'insertion supérieur de 1.9653 % à ceux de la discipline de référence.

La valeur p (0.000) indique également une forte significativité statistique.

Informatique : 5.9681

Les diplômés en Informatique ont un taux d'insertion supérieur de 5.9681 % par rapport à la discipline de référence.

Cet effet est très significatif avec une valeur p proche de zéro.

Histoire-géographie : -5.3506

Les diplômés en Histoire-géographie ont un taux d'insertion inférieur de 5.3506 % par rapport à la discipline de référence.

La valeur p (0.000) montre que cet effet est statistiquement significatif.

Analyse des autres résultats

Ensemble sciences humaines et sociales (-3.1474) et Lettres, langues, arts (-4.5183) montrent des coefficients négatifs, indiquant que ces disciplines sont associées à un taux d'insertion inférieur par rapport à la discipline de référence.

Masters enseignement (7.6375), Masters enseignement : second degré, CPE... (7.3016) et Masters enseignement : premier degré (6.2082) ont des coefficients élevés et positifs, ce qui suggère que les diplômés de ces disciplines ont un taux d'insertion nettement supérieur.

Sciences de la vie et de la terre (-4.2667) et Psychologie (1.6673) sont également significatifs, mais avec des effets différents sur le taux d'insertion.

Signification des p-values

Les p-values indiquent la significativité statistique des coefficients. Une valeur p inférieure à 0.05 (ou à 0.01 dans de nombreux cas) suggère que l'effet observé est statistiquement significatif.

Par exemple, tous les coefficients des disciplines sont statistiquement significatifs, à l'exception de Sciences fondamentales (-0.5879) et Économie (0.0223), qui ne sont pas significatifs au niveau de 5 % (les p-values sont respectivement 0.235 et 0.963).

```
plt.figure(figsize=(14,8))

sns.barplot(x='discipline', y='taux_dinsertion', data=df,
palette='viridis')

plt.title("Taux d'Insertion par Discipline", fontsize=16)
plt.xlabel("Discipline", fontsize=14)
plt.ylabel("Taux d'Insertion", fontsize=14)

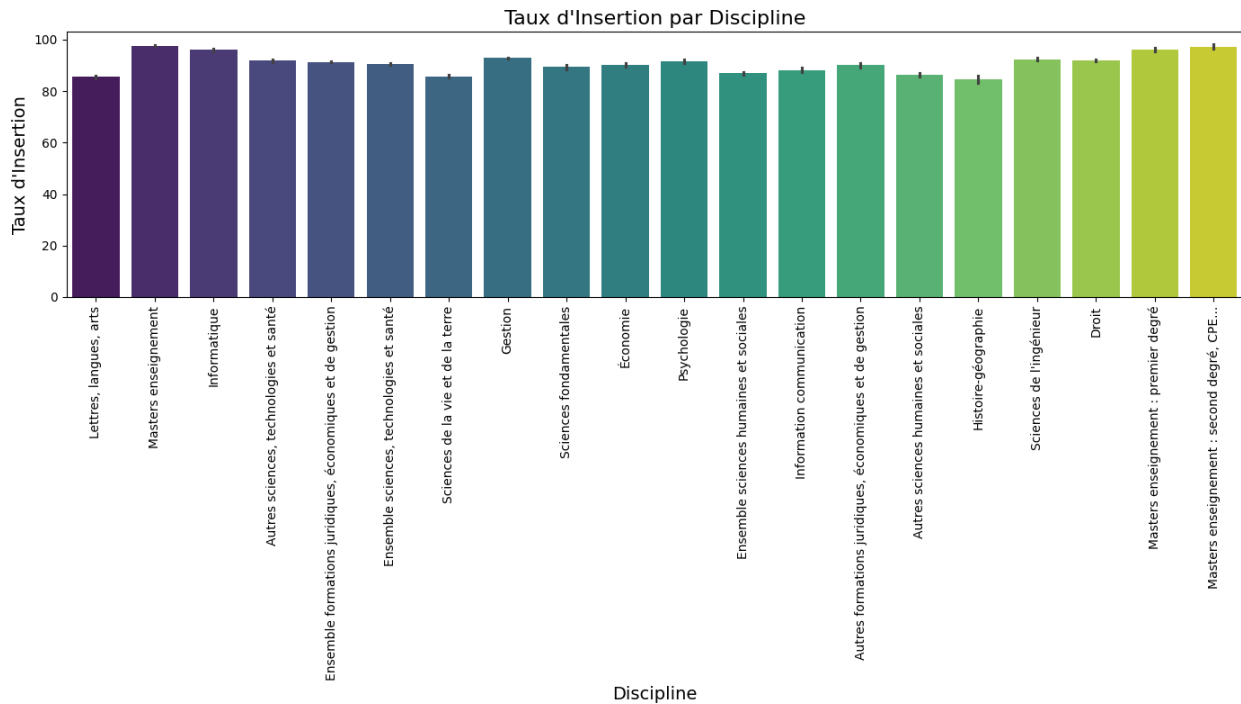
plt.xticks(rotation=90)

file_path =
'/content/drive/MyDrive/graphique_insertion_par_discipline.png'
plt.tight_layout()
plt.savefig(file_path, dpi=300)

plt.show()

<ipython-input-268-80acc5d08b93>:3: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be
removed in v0.14.0. Assign the `x` variable to `hue` and set
`legend=False` for the same effect.

sns.barplot(x='discipline', y='taux_dinsertion', data=df,
palette='viridis')
```



```

groupby_1 = df.groupby('discipline')['emplois_stables'].mean()
groupby_sorted3 = groupby_1.sort_values(ascending=True)

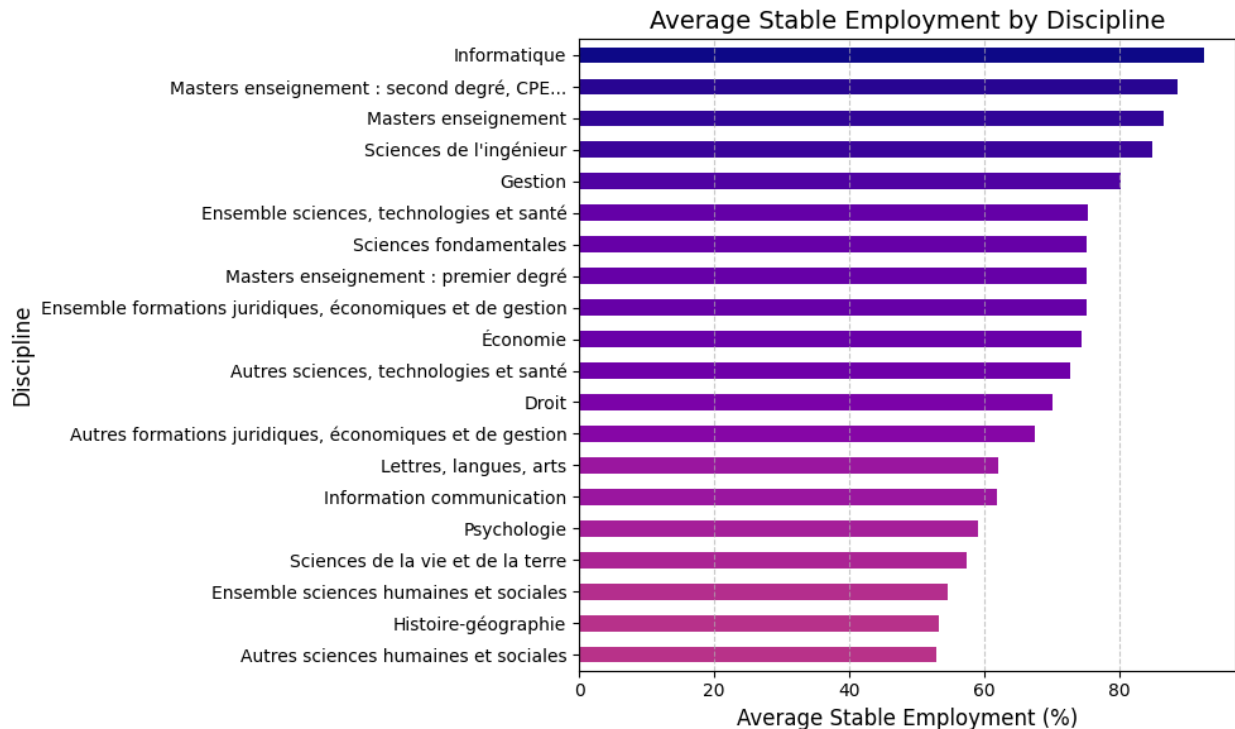
plt.figure(figsize=(10, 6))
groupby_sorted3.plot(kind='barh',
                      color=plt.cm.plasma(1 - (groupby_sorted3.values /
max(groupby_sorted3.values))))

plt.title("Average Stable Employment by Discipline", fontsize=14)
plt.xlabel("Average Stable Employment (%)", fontsize=12)
plt.ylabel("Discipline", fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)

file_path = '/content/drive/MyDrive/'
plt.tight_layout()
plt.savefig(file_path, dpi=300)

plt.tight_layout()
plt.show()

```



Regression_multiple

```
y = df['taux_dinsertion']
x1 = df['discipline']
x2 = df['etablissement']
multivariate_regression = smf.ols(formula=' y ~ x1 + x2 ',
data=df).fit()
```

```
print(multivariate_regression.summary())
```

OLS Regression Results

```
=====
=====
Dep. Variable:                y    R-squared:
0.410
Model:                        OLS    Adj. R-squared:
0.402
Method:                       Least Squares    F-statistic:
49.80
Date:                          Mon, 18 Nov 2024    Prob (F-statistic):
0.00
Time:                          16:18:03    Log-Likelihood:
-21407.
No. Observations:              7346    AIC:
4.302e+04
Df Residuals:                  7244    BIC:
```


4.372e+04

Df Model: 101

Covariance Type: nonrobust

coef	std err	t	P> t	[0.025	0.975]

Intercept					
86.7446	0.454	191.029	0.000	85.854	87.635
x1[T.Autres sciences humaines et sociales]					-
2.6449	0.453	-5.835	0.000	-3.533	-1.756
x1[T.Autres sciences, technologies et santé]					
2.0307	0.484	4.196	0.000	1.082	2.980
x1[T.Droit]					
2.1052	0.422	4.994	0.000	1.279	2.932
x1[T.Ensemble formations juridiques, économiques et de gestion]					
1.5251	0.392	3.891	0.000	0.757	2.293
x1[T.Ensemble sciences humaines et sociales]					-
2.6990	0.402	-6.716	0.000	-3.487	-1.911
x1[T.Ensemble sciences, technologies et santé]					
0.5354	0.399	1.343	0.179	-0.246	1.317
x1[T.Gestion]					
2.9265	0.398	7.361	0.000	2.147	3.706
x1[T.Histoire-géographie]					-
4.4042	0.609	-7.237	0.000	-5.597	-3.211
x1[T.Information communication]					-
1.0917	0.497	-2.198	0.028	-2.065	-0.118
x1[T.Informatique]					
6.1067	0.462	13.209	0.000	5.200	7.013
x1[T.Lettres, langues, arts]					-
3.6359	0.437	-8.328	0.000	-4.492	-2.780
x1[T.Masters enseignement]					
8.0151	0.412	19.472	0.000	7.208	8.822
x1[T.Masters enseignement : premier degré]					
6.5554	0.929	7.060	0.000	4.735	8.376
x1[T.Masters enseignement : second degré, CPE...]					
8.1863	0.956	8.561	0.000	6.312	10.061
x1[T.Psychologie]					
2.3137	0.494	4.684	0.000	1.345	3.282
x1[T.Sciences de l'ingénieur]					
2.5343	0.459	5.517	0.000	1.634	3.435
x1[T.Sciences de la vie et de la terre]					-
3.9315	0.437	-9.005	0.000	-4.787	-3.076
x1[T.Sciences fondamentales]					-
0.0002	0.489	-0.000	1.000	-0.958	0.958
x1[T.Économie]					

0.4225	0.465	0.908	0.364	-0.490	1.335
x2[T.Amiens - Picardie Jules-Verne]					
4.8761	0.482	10.119	0.000	3.932	5.821
x2[T.Angers]					
2.8842	0.469	6.153	0.000	1.965	3.803
x2[T.Artois]					
3.8514	0.574	6.709	0.000	2.726	4.977
x2[T.Avignon]					
1.9921	1.047	1.903	0.057	-0.060	4.044
x2[T.Besançon - Franche-Comté]					
2.4387	0.519	4.701	0.000	1.422	3.456
x2[T.Bordeaux]					
2.8216	0.403	6.998	0.000	2.031	3.612
x2[T.Bordeaux - Montaigne]					
0.6347	0.558	1.136	0.256	-0.460	1.730
x2[T.Brest - Bretagne occidentale]					
1.4296	0.451	3.173	0.002	0.546	2.313
x2[T.Bretagne-Sud]					
5.2746	0.563	9.375	0.000	4.172	6.378
x2[T.CY Cergy Paris]					
5.4948	0.942	5.836	0.000	3.649	7.340
x2[T.Caen Normandie]					
3.9583	0.457	8.665	0.000	3.063	4.854
x2[T.Cergy-Pontoise]					
5.3093	0.527	10.073	0.000	4.276	6.343
x2[T.Clermont Auvergne]					
4.1220	0.482	8.548	0.000	3.177	5.067
x2[T.Clermont-Ferrand 1 - Auvergne]					
5.2790	1.100	4.801	0.000	3.124	7.434
x2[T.Clermont-Ferrand 2 - Blaise Pascal]					
2.1189	0.880	2.409	0.016	0.395	3.843
x2[T.Corse Pasquale Paoli]					
0.5988	1.615	0.371	0.711	-2.567	3.764
x2[T.Côte d'Azur]					
3.6831	0.877	4.199	0.000	1.963	5.403
x2[T.Dijon - Bourgogne]					
2.6618	0.471	5.651	0.000	1.738	3.585
x2[T.Grenoble 1 - Joseph Fourier]					
1.5105	1.096	1.379	0.168	-0.637	3.658
x2[T.Grenoble 2 - Pierre Mendès France]					-
0.1631	1.157	-0.141	0.888	-2.431	2.105
x2[T.Grenoble 3 - Stendhal]					-
1.2691	1.527	-0.831	0.406	-4.263	1.725
x2[T.Grenoble Alpes]					
2.7574	0.411	6.707	0.000	1.951	3.563
x2[T.Gustave Eiffel]					
2.3125	1.232	1.878	0.060	-0.102	4.727
x2[T.La Rochelle]					
3.9799	0.601	6.619	0.000	2.801	5.159

x2[T.La Réunion]						-
0.7597	4.504	-0.169	0.866	-9.588	8.069	
x2[T.Le Havre]						-
0.5204	0.741	-0.702	0.483	-1.973	0.933	
x2[T.Le Mans]						
3.5588	0.765	4.649	0.000	2.058	5.059	
x2[T.Lille]						
4.3703	0.428	10.215	0.000	3.532	5.209	
x2[T.Lille 1 - Sciences technologies]						
1.5978	0.589	2.713	0.007	0.443	2.752	
x2[T.Lille 2 - Droit et santé]						
4.9736	0.810	6.143	0.000	3.387	6.561	
x2[T.Lille 3 - Charles-de-Gaulle]						
2.0836	0.801	2.601	0.009	0.513	3.654	
x2[T.Lille Nord de France]						
2.7403	1.863	1.471	0.141	-0.912	6.392	
x2[T.Limoges]						
4.7211	0.649	7.278	0.000	3.449	5.993	
x2[T.Littoral - Côte d'Opale]						
1.8827	0.615	3.061	0.002	0.677	3.088	
x2[T.Lorraine]						
3.4274	0.399	8.579	0.000	2.644	4.210	
x2[T.Lyon 1 - Claude Bernard]						
3.0814	0.496	6.209	0.000	2.109	4.054	
x2[T.Lyon 2 - Lumière]						
3.6137	0.420	8.611	0.000	2.791	4.436	
x2[T.Lyon 3 - Jean Moulin]						
1.7938	0.471	3.807	0.000	0.870	2.717	
x2[T.Montpellier]						
2.0168	0.448	4.499	0.000	1.138	2.896	
x2[T.Montpellier 1]						
2.6232	2.028	1.293	0.196	-1.352	6.599	
x2[T.Montpellier 2 - Sciences techniques du Languedoc]						
0.1555	1.725	0.090	0.928	-3.226	3.537	
x2[T.Montpellier 3 - Paul-Valéry]						-
2.5187	0.532	-4.735	0.000	-3.561	-1.476	
x2[T.Mulhouse - Haute Alsace]						
3.5126	0.695	5.053	0.000	2.150	4.875	
x2[T.Nantes]						
3.4494	0.447	7.716	0.000	2.573	4.326	
x2[T.Nice - Sophia-Antipolis]						
1.4794	0.495	2.990	0.003	0.510	2.449	
x2[T.Orléans]						
2.9264	0.540	5.419	0.000	1.868	3.985	
x2[T.Paris 1 - Panthéon Sorbonne]						
3.1023	0.449	6.908	0.000	2.222	3.983	
x2[T.Paris 11 - Paris Sud]						
2.1642	0.671	3.224	0.001	0.848	3.480	
x2[T.Paris 2 - Panthéon Assas]						

5.0523	0.653	7.733	0.000	3.772	6.333	
x2[T.Paris 3 - Sorbonne Nouvelle]						
2.8120	0.668	4.208	0.000	1.502	4.122	
x2[T.Paris 4 - Sorbonne]						
5.3532	0.870	6.151	0.000	3.647	7.059	
x2[T.Paris 7 - Denis Diderot]						
4.0349	0.693	5.823	0.000	2.677	5.393	
x2[T.Paris 8 - Vincennes Saint-Denis]						
2.8186	0.655	4.303	0.000	1.535	4.103	
x2[T.Paris Nanterre]						
3.7665	0.501	7.522	0.000	2.785	4.748	
x2[T.Paris Pierre et Marie Curie]						-
1.4450	0.792	-1.824	0.068	-2.998	0.108	
x2[T.Paris René Descartes]						
4.1058	0.607	6.759	0.000	2.915	5.297	
x2[T.Paris-Est Créteil]						
3.9023	0.486	8.022	0.000	2.949	4.856	
x2[T.Paris-Est Marne-la-Vallée]						
3.6816	0.522	7.046	0.000	2.657	4.706	
x2[T.Paris-Panthéon-Assas]						
5.5545	2.029	2.738	0.006	1.578	9.531	
x2[T.Paris-Saclay]						
4.6085	0.523	8.812	0.000	3.583	5.634	
x2[T.Pau - Pays de l'Adour]						
2.7829	0.588	4.731	0.000	1.630	3.936	
x2[T.Perpignan - Via Domitia]						
3.2522	0.735	4.423	0.000	1.811	4.694	
x2[T.Poitiers]						
3.2742	0.474	6.901	0.000	2.344	4.204	
x2[T.Polytechnique Hauts-de-France]						
5.9585	0.978	6.095	0.000	4.042	7.875	
x2[T.Reims - Champagne Ardenne]						
3.1945	0.525	6.088	0.000	2.166	4.223	
x2[T.Rennes 1]						
2.1879	0.477	4.583	0.000	1.252	3.124	
x2[T.Rennes 2]						
0.7796	0.501	1.556	0.120	-0.202	1.761	
x2[T.Rouen]						
3.6791	0.478	7.704	0.000	2.743	4.615	
x2[T.Saint-Etienne - Jean Monnet]						
4.4058	0.570	7.730	0.000	3.289	5.523	
x2[T.Savoie Mont Blanc]						
4.4120	0.529	8.343	0.000	3.375	5.449	
x2[T.Sorbonne Université]						
3.1979	0.512	6.246	0.000	2.194	4.201	
x2[T.Strasbourg]						
3.0309	0.411	7.383	0.000	2.226	3.836	
x2[T.Toulon]						
0.8304	0.626	1.327	0.185	-0.396	2.057	

x2[T.Toulouse - Jean Jaures]					-
0.5332	0.517	-1.031	0.303	-1.547	0.481
x2[T.Toulouse 1 - Capitole]					
4.8721	0.518	9.397	0.000	3.856	5.888
x2[T.Toulouse 3 - Paul Sabatier]					
4.6844	0.488	9.608	0.000	3.729	5.640
x2[T.Tours]					
3.1237	0.503	6.205	0.000	2.137	4.111
x2[T.Université Polytechnique Hauts-de-France]					
4.7156	0.608	7.757	0.000	3.524	5.907
x2[T.Université Sorbonne Paris Nord]					
3.8100	0.702	5.429	0.000	2.434	5.186
x2[T.Université de Paris]					
5.4978	0.557	9.869	0.000	4.406	6.590
x2[T.Versailles Saint-Quentin-en-Yvelines]					
2.4661	0.560	4.404	0.000	1.368	3.564
x2[T.Évry-Val d'Essonne]					-
0.9122	0.608	-1.500	0.134	-2.104	0.280

```
=====
=====
Omnibus:                638.285    Durbin-Watson:
1.808
Prob(Omnibus):          0.000    Jarque-Bera (JB):
1016.627
Skew:                   -0.654    Prob(JB):
1.75e-221
Kurtosis:               4.268    Cond. No.
90.0
=====
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
y = df['emplois_stables']
x1 = df['discipline']
x2 = df['etablissement']
x3 = df['emplois_cadre_ou_professions_intermediaires']
x4 = df['salaire_brut_annuel_estime']
x5 = df['emplois_a_temps_plein']
x5 = df['emplois_cadre']
multivariate_regression = smf.ols(formula=' y ~ x1 + x2 + x3 + x4 +
x5', data=df).fit()
```

```
print(multivariate_regression.summary())
```

OLS Regression Results

```

=====
=====
Dep. Variable:                y    R-squared:
0.717
Model:                        OLS    Adj. R-squared:
0.713
Method:                        Least Squares    F-statistic:
176.2
Date:                          Mon, 18 Nov 2024    Prob (F-statistic):
0.00
Time:                          16:18:03    Log-Likelihood:
-25906.
No. Observations:              7346    AIC:
5.202e+04
Df Residuals:                  7241    BIC:
5.275e+04
Df Model:                      104

```

Covariance Type: nonrobust

```

=====
=====
coef      std err          t      P>|t|      [0.025      0.975]
-----
-----
Intercept
7.1280      1.683      -4.235      0.000     -10.427      -3.828
x1[T.Autres sciences humaines et sociales]
5.3833      0.850      -6.336      0.000      -7.049      -3.718
x1[T.Autres sciences, technologies et santé]
6.5761      0.896       7.341      0.000       4.820       8.332
x1[T.Droit]
2.8850      0.779       3.705      0.000       1.359       4.411
x1[T.Ensemble formations juridiques, économiques et de gestion]
5.9628      0.724       8.234      0.000       4.543       7.382
x1[T.Ensemble sciences humaines et sociales]
4.3922      0.766      -5.731      0.000      -5.895      -2.890
x1[T.Ensemble sciences, technologies et santé]
3.9146      0.749       5.224      0.000       2.446       5.384
x1[T.Gestion]
9.2082      0.737      12.493      0.000       7.763      10.653
x1[T.Histoire-géographie]
5.1126      1.133      -4.513      0.000      -7.333      -2.892
x1[T.Information communication]
1.2905      0.927       1.393      0.164      -0.526       3.107
x1[T.Informatique]
14.1253      0.900      15.689      0.000      12.360      15.890
x1[T.Lettres, langues, arts]
7.5395      0.831       9.067      0.000       5.910       9.169

```

x1[T.Masters enseignement]					
23.4578	0.843	27.823	0.000	21.805	25.111
x1[T.Masters enseignement : premier degré]					
15.9497	1.744	9.148	0.000	12.532	19.368
x1[T.Masters enseignement : second degré, CPE...]					
23.9582	1.802	13.297	0.000	20.426	27.490
x1[T.Psychologie]					-
2.5170	0.995	-2.530	0.011	-4.467	-0.567
x1[T.Sciences de l'ingénieur]					
9.1218	0.871	10.467	0.000	7.414	10.830
x1[T.Sciences de la vie et de la terre]					-
6.3955	0.818	-7.816	0.000	-7.999	-4.791
x1[T.Sciences fondamentales]					
1.4288	0.916	1.560	0.119	-0.367	3.224
x1[T.Économie]					
3.0186	0.864	3.495	0.000	1.326	4.712
x2[T.Amiens - Picardie Jules-Verne]					
4.2861	0.890	4.813	0.000	2.541	6.032
x2[T.Angers]					
0.9129	0.870	1.049	0.294	-0.793	2.619
x2[T.Artois]					
6.7044	1.064	6.304	0.000	4.619	8.789
x2[T.Avignon]					
7.7678	1.935	4.015	0.000	3.975	11.561
x2[T.Besançon - Franche-Comté]					
4.5992	0.958	4.801	0.000	2.721	6.477
x2[T.Bordeaux]					
0.4942	0.749	0.660	0.509	-0.973	1.962
x2[T.Bordeaux - Montaigne]					-
3.3260	1.037	-3.208	0.001	-5.358	-1.294
x2[T.Brest - Bretagne occidentale]					-
2.7528	0.839	-3.282	0.001	-4.397	-1.109
x2[T.Bretagne-Sud]					
7.7936	1.041	7.489	0.000	5.754	9.834
x2[T.CY Cergy Paris]					-
2.4216	1.759	-1.377	0.169	-5.870	1.027
x2[T.Caen Normandie]					-
1.4156	0.845	-1.676	0.094	-3.071	0.240
x2[T.Cergy-Pontoise]					
0.9379	0.992	0.945	0.344	-1.007	2.883
x2[T.Clermont Auvergne]					
2.0716	0.892	2.323	0.020	0.323	3.820
x2[T.Clermont-Ferrand 1 - Auvergne]					
8.4692	2.031	4.171	0.000	4.489	12.450
x2[T.Clermont-Ferrand 2 - Blaise Pascal]					
2.1737	1.625	1.337	0.181	-1.012	5.360
x2[T.Corse Pasquale Paoli]					-
3.0594	2.989	-1.024	0.306	-8.919	2.800
x2[T.Côte d'Azur]					

3.5814	1.626	2.202	0.028	0.393	6.770	
x2[T.Dijon - Bourgogne]						
2.5176	0.871	2.889	0.004	0.809	4.226	
x2[T.Grenoble 1 - Joseph Fourier]						
2.3598	2.023	1.167	0.243	-1.605	6.325	
x2[T.Grenoble 2 - Pierre Mendès France]						-
1.4404	2.139	-0.673	0.501	-5.634	2.753	
x2[T.Grenoble 3 - Stendhal]						-
7.7228	2.819	-2.740	0.006	-13.249	-2.197	
x2[T.Grenoble Alpes]						-
1.2763	0.764	-1.672	0.095	-2.773	0.220	
x2[T.Gustave Eiffel]						-
1.1169	2.282	-0.489	0.625	-5.590	3.356	
x2[T.La Rochelle]						
7.9982	1.116	7.169	0.000	5.811	10.185	
x2[T.La Réunion]						-
31.6426	8.332	-3.798	0.000	-47.975	-15.310	
x2[T.Le Havre]						
1.2952	1.377	0.941	0.347	-1.404	3.994	
x2[T.Le Mans]						
6.3338	1.417	4.470	0.000	3.556	9.112	
x2[T.Lille]						
2.0073	0.797	2.518	0.012	0.445	3.570	
x2[T.Lille 1 - Sciences technologies]						-
0.1028	1.096	-0.094	0.925	-2.252	2.047	
x2[T.Lille 2 - Droit et santé]						
6.1682	1.498	4.118	0.000	3.232	9.104	
x2[T.Lille 3 - Charles-de-Gaulle]						
0.2183	1.480	0.147	0.883	-2.684	3.120	
x2[T.Lille Nord de France]						
3.6071	3.440	1.049	0.294	-3.136	10.350	
x2[T.Limoges]						
3.1597	1.198	2.636	0.008	0.810	5.509	
x2[T.Littoral - Côte d'Opale]						
5.2057	1.141	4.561	0.000	2.968	7.443	
x2[T.Lorraine]						-
0.5610	0.740	-0.758	0.449	-2.012	0.890	
x2[T.Lyon 1 - Claude Bernard]						
1.1674	0.918	1.272	0.203	-0.632	2.967	
x2[T.Lyon 2 - Lumière]						
2.1920	0.775	2.827	0.005	0.672	3.712	
x2[T.Lyon 3 - Jean Moulin]						
0.2569	0.883	0.291	0.771	-1.473	1.987	
x2[T.Montpellier]						
0.0867	0.831	0.104	0.917	-1.543	1.716	
x2[T.Montpellier 1]						
6.9241	3.744	1.849	0.064	-0.415	14.263	
x2[T.Montpellier 2 - Sciences techniques du Languedoc]						
4.8360	3.184	1.519	0.129	-1.406	11.078	

x2[T.Montpellier 3 - Paul-Valéry]						-
0.6120	0.984	-0.622	0.534	-2.541	1.317	
x2[T.Mulhouse - Haute Alsace]						
1.2511	1.284	0.974	0.330	-1.266	3.768	
x2[T.Nantes]						-
1.0700	0.827	-1.293	0.196	-2.692	0.552	
x2[T.Nice - Sophia-Antipolis]						
5.1860	0.914	5.677	0.000	3.395	6.977	
x2[T.Orléans]						
0.7775	1.002	0.776	0.438	-1.188	2.743	
x2[T.Paris 1 - Panthéon Sorbonne]						-
11.0587	0.853	-12.957	0.000	-12.732	-9.386	
x2[T.Paris 11 - Paris Sud]						-
0.2566	1.245	-0.206	0.837	-2.698	2.184	
x2[T.Paris 2 - Panthéon Assas]						-
12.8149	1.250	-10.252	0.000	-15.265	-10.365	
x2[T.Paris 3 - Sorbonne Nouvelle]						-
3.7759	1.234	-3.060	0.002	-6.195	-1.357	
x2[T.Paris 4 - Sorbonne]						
2.8841	1.618	1.782	0.075	-0.289	6.057	
x2[T.Paris 7 - Denis Diderot]						-
0.9640	1.289	-0.748	0.454	-3.490	1.562	
x2[T.Paris 8 - Vincennes Saint-Denis]						-
3.6902	1.214	-3.040	0.002	-6.070	-1.311	
x2[T.Paris Nanterre]						
2.2589	0.936	2.412	0.016	0.423	4.095	
x2[T.Paris Pierre et Marie Curie]						-
1.3570	1.468	-0.925	0.355	-4.234	1.520	
x2[T.Paris René Descartes]						-
2.2959	1.132	-2.027	0.043	-4.516	-0.076	
x2[T.Paris-Est Créteil]						-
1.3268	0.909	-1.460	0.144	-3.108	0.454	
x2[T.Paris-Est Marne-la-Vallée]						
3.2392	0.974	3.326	0.001	1.330	5.148	
x2[T.Paris-Panthéon-Assas]						-
12.7606	3.775	-3.380	0.001	-20.161	-5.360	
x2[T.Paris-Saclay]						-
5.2543	0.995	-5.283	0.000	-7.204	-3.305	
x2[T.Pau - Pays de l'Adour]						
2.1369	1.091	1.958	0.050	-0.002	4.276	
x2[T.Perpignan - Via Domitia]						
9.9751	1.372	7.273	0.000	7.286	12.664	
x2[T.Poitiers]						
1.1061	0.876	1.263	0.207	-0.610	2.823	
x2[T.Polytechnique Hauts-de-France]						
5.2991	1.808	2.931	0.003	1.755	8.844	
x2[T.Reims - Champagne Ardenne]						
3.5863	0.970	3.696	0.000	1.684	5.488	
x2[T.Rennes 1]						

0.3655	0.886	0.413	0.680	-1.371	2.102	
x2[T.Rennes 2]						-
3.6785	0.926	-3.972	0.000	-5.494	-1.863	
x2[T.Rouen]						
0.4304	0.881	0.488	0.625	-1.297	2.158	
x2[T.Saint-Etienne - Jean Monnet]						
2.7866	1.059	2.632	0.009	0.711	4.862	
x2[T.Savoie Mont Blanc]						
6.1119	0.978	6.251	0.000	4.195	8.029	
x2[T.Sorbonne Université]						-
3.2340	0.969	-3.339	0.001	-5.133	-1.335	
x2[T.Strasbourg]						-
3.3094	0.760	-4.355	0.000	-4.799	-1.820	
x2[T.Toulon]						
14.8380	1.168	12.701	0.000	12.548	17.128	
x2[T.Toulouse - Jean Jaures]						
0.7644	0.957	0.799	0.424	-1.111	2.640	
x2[T.Toulouse 1 - Capitole]						
5.2371	0.964	5.435	0.000	3.348	7.126	
x2[T.Toulouse 3 - Paul Sabatier]						
6.4411	0.908	7.092	0.000	4.661	8.221	
x2[T.Tours]						-
2.2789	0.934	-2.439	0.015	-4.110	-0.447	
x2[T.Université Polytechnique Hauts-de-France]						
5.3454	1.132	4.723	0.000	3.127	7.564	
x2[T.Université Sorbonne Paris Nord]						
2.1271	1.301	1.635	0.102	-0.423	4.677	
x2[T.Université de Paris]						-
7.1710	1.056	-6.788	0.000	-9.242	-5.100	
x2[T.Versailles Saint-Quentin-en-Yvelines]						-
4.7540	1.041	-4.569	0.000	-6.794	-2.714	
x2[T.Évry-Val d'Essonne]						-
1.4412	1.127	-1.279	0.201	-3.650	0.768	
x3						
0.1259	0.018	7.071	0.000	0.091	0.161	
x4						
0.0020	4.74e-05	42.740	0.000	0.002	0.002	
x5						
0.0230	0.012	1.874	0.061	-0.001	0.047	
=====						
=====						
Omnibus:		144.667		Durbin-Watson:		
1.900						
Prob(Omnibus):		0.000		Jarque-Bera (JB):		
185.217						
Skew:		-0.262		Prob(JB):		
6.04e-41						
Kurtosis:		3.574		Cond. No.		
2.60e+06						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.6e+06. This might indicate that there are strong multicollinearity or other numerical problems.

```
y = df['emplois_stables']
```

```
x1 = df['discipline']
```

```
x2 = df['etablissement']
```

```
multivariate_regression = smf.ols(formula=' y ~ x1 + x2 ',  
data=df).fit()
```

```
print(multivariate_regression.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          y      R-squared:
0.582
Model:                  OLS    Adj. R-squared:
0.577
Method:                 Least Squares    F-statistic:
100.0
Date:                   Mon, 18 Nov 2024    Prob (F-statistic):
0.00
Time:                   16:18:03    Log-Likelihood:
-27334.
No. Observations:       7346    AIC:
5.487e+04
Df Residuals:           7244    BIC:
5.558e+04
Df Model:               101
```

```
Covariance Type:        nonrobust
```

```
=====
=====
coef      std err        t    P>|t|    [0.025    0.975]
-----
-----
Intercept
64.8479    1.017    63.736    0.000    62.853    66.842
x1[T.Autres sciences humaines et sociales] -
```

13.2549	1.016	-13.051	0.000	-15.246	-11.264
x1[T.Autres sciences, technologies et santé]					
5.2844	1.084	4.873	0.000	3.158	7.410
x1[T.Droit]					
2.2770	0.945	2.410	0.016	0.425	4.129
x1[T.Ensemble formations juridiques, économiques et de gestion]					
7.2689	0.878	8.277	0.000	5.547	8.991
x1[T.Ensemble sciences humaines et sociales]					-
12.7852	0.900	-14.200	0.000	-14.550	-11.020
x1[T.Ensemble sciences, technologies et santé]					
7.6858	0.893	8.606	0.000	5.935	9.436
x1[T.Gestion]					
12.7315	0.891	14.293	0.000	10.985	14.478
x1[T.Histoire-géographie]					-
13.7681	1.364	-10.097	0.000	-16.441	-11.095
x1[T.Information communication]					-
6.0263	1.113	-5.415	0.000	-8.208	-3.845
x1[T.Informatique]					
24.1200	1.036	23.285	0.000	22.089	26.151
x1[T.Lettres, langues, arts]					-
4.1986	0.978	-4.292	0.000	-6.116	-2.281
x1[T.Masters enseignement]					
19.4794	0.922	21.121	0.000	17.671	21.287
x1[T.Masters enseignement : premier degré]					
8.1860	2.080	3.935	0.000	4.108	12.264
x1[T.Masters enseignement : second degré, CPE...]					
22.4229	2.143	10.465	0.000	18.223	26.623
x1[T.Psychologie]					-
7.5195	1.107	-6.794	0.000	-9.689	-5.350
x1[T.Sciences de l'ingénieur]					
16.1937	1.029	15.734	0.000	14.176	18.211
x1[T.Sciences de la vie et de la terre]					-
9.4688	0.978	-9.680	0.000	-11.386	-7.551
x1[T.Sciences fondamentales]					
7.6399	1.095	6.979	0.000	5.494	9.786
x1[T.Économie]					
7.8049	1.043	7.486	0.000	5.761	9.849
x2[T.Amiens - Picardie Jules-Verne]					
5.8468	1.080	5.415	0.000	3.730	7.963
x2[T.Angers]					-
3.0786	1.050	-2.931	0.003	-5.138	-1.020
x2[T.Artois]					
6.1038	1.286	4.746	0.000	3.582	8.625
x2[T.Avignon]					
7.7805	2.345	3.317	0.001	3.183	12.378
x2[T.Besançon - Franche-Comté]					
2.4707	1.162	2.126	0.034	0.192	4.749
x2[T.Bordeaux]					
1.5940	0.903	1.764	0.078	-0.177	3.365

x2[T.Bordeaux - Montaigne]						-
3.0371	1.251	-2.427	0.015	-5.490	-0.584	
x2[T.Brest - Bretagne occidentale]						-
4.5961	1.010	-4.552	0.000	-6.575	-2.617	
x2[T.Bretagne-Sud]						
6.1200	1.261	4.855	0.000	3.649	8.591	
x2[T.CY Cergy Paris]						
12.9187	2.110	6.124	0.000	8.783	17.054	
x2[T.Caen Normandie]						-
0.3662	1.024	-0.358	0.721	-2.373	1.640	
x2[T.Cergy-Pontoise]						
11.9228	1.181	10.095	0.000	9.608	14.238	
x2[T.Clermont Auvergne]						
0.0861	1.081	0.080	0.936	-2.032	2.204	
x2[T.Clermont-Ferrand 1 - Auvergne]						
6.7725	2.464	2.749	0.006	1.943	11.602	
x2[T.Clermont-Ferrand 2 - Blaise Pascal]						-
2.6094	1.971	-1.324	0.186	-6.473	1.254	
x2[T.Corse Pasquale Paoli]						-
15.2524	3.618	-4.216	0.000	-22.345	-8.160	
x2[T.Côte d'Azur]						
10.9057	1.966	5.548	0.000	7.053	14.759	
x2[T.Dijon - Bourgogne]						
0.8808	1.055	0.835	0.404	-1.188	2.950	
x2[T.Grenoble 1 - Joseph Fourier]						
1.7318	2.455	0.706	0.481	-3.080	6.544	
x2[T.Grenoble 2 - Pierre Mendès France]						-
1.9462	2.593	-0.751	0.453	-7.028	3.136	
x2[T.Grenoble 3 - Stendhal]						-
10.4001	3.422	-3.039	0.002	-17.108	-3.692	
x2[T.Grenoble Alpes]						
1.0045	0.921	1.090	0.276	-0.801	2.810	
x2[T.Gustave Eiffel]						
9.8457	2.760	3.568	0.000	4.436	15.256	
x2[T.La Rochelle]						
6.2879	1.347	4.667	0.000	3.647	8.929	
x2[T.La Réunion]						-
3.3273	10.091	-0.330	0.742	-23.108	16.453	
x2[T.Le Havre]						-
3.2009	1.661	-1.927	0.054	-6.457	0.055	
x2[T.Le Mans]						
1.0534	1.715	0.614	0.539	-2.309	4.416	
x2[T.Lille]						
5.1896	0.959	5.414	0.000	3.310	7.069	
x2[T.Lille 1 - Sciences technologies]						-
1.6643	1.320	-1.261	0.207	-4.251	0.923	
x2[T.Lille 2 - Droit et santé]						
6.3883	1.814	3.522	0.000	2.832	9.944	
x2[T.Lille 3 - Charles-de-Gaulle]						-

1.2536	1.795	-0.698	0.485	-4.772	2.265	
x2[T.Lille Nord de France]						
6.0060	4.174	1.439	0.150	-2.177	14.189	
x2[T.Limoges]						
1.9571	1.453	1.347	0.178	-0.892	4.806	
x2[T.Littoral - Côte d'Opale]						
3.6385	1.378	2.640	0.008	0.937	6.340	
x2[T.Lorraine]						
3.3162	0.895	3.705	0.000	1.562	5.071	
x2[T.Lyon 1 - Claude Bernard]						
2.9205	1.112	2.627	0.009	0.741	5.100	
x2[T.Lyon 2 - Lumière]						
0.6934	0.940	0.737	0.461	-1.150	2.537	
x2[T.Lyon 3 - Jean Moulin]						
4.7185	1.056	4.470	0.000	2.649	6.788	
x2[T.Montpellier]						-
0.0093	1.004	-0.009	0.993	-1.978	1.960	
x2[T.Montpellier 1]						
11.4532	4.544	2.520	0.012	2.546	20.361	
x2[T.Montpellier 2 - Sciences techniques du Languedoc]						
5.6105	3.865	1.452	0.147	-1.966	13.186	
x2[T.Montpellier 3 - Paul-Valéry]						-
4.0955	1.192	-3.436	0.001	-6.432	-1.759	
x2[T.Mulhouse - Haute Alsace]						
2.6368	1.557	1.693	0.090	-0.416	5.690	
x2[T.Nantes]						
0.0367	1.002	0.037	0.971	-1.927	2.000	
x2[T.Nice - Sophia-Antipolis]						
5.3376	1.108	4.815	0.000	3.165	7.510	
x2[T.Orléans]						
2.3047	1.210	1.905	0.057	-0.067	4.677	
x2[T.Paris 1 - Panthéon Sorbonne]						-
0.2359	1.006	-0.234	0.815	-2.208	1.737	
x2[T.Paris 11 - Paris Sud]						
6.4605	1.504	4.295	0.000	3.512	9.409	
x2[T.Paris 2 - Panthéon Assas]						
4.8812	1.464	3.334	0.001	2.012	7.751	
x2[T.Paris 3 - Sorbonne Nouvelle]						-
1.6060	1.497	-1.073	0.284	-4.541	1.329	
x2[T.Paris 4 - Sorbonne]						
11.9039	1.950	6.104	0.000	8.081	15.727	
x2[T.Paris 7 - Denis Diderot]						
6.2364	1.553	4.017	0.000	3.193	9.280	
x2[T.Paris 8 - Vincennes Saint-Denis]						
1.1085	1.468	0.755	0.450	-1.768	3.985	
x2[T.Paris Nanterre]						
9.4104	1.122	8.388	0.000	7.211	11.610	
x2[T.Paris Pierre et Marie Curie]						
3.9873	1.775	2.246	0.025	0.508	7.467	

x2[T.Paris René Descartes]					
6.4491	1.361	4.738	0.000	3.781	9.117
x2[T.Paris-Est Créteil]					
5.6048	1.090	5.142	0.000	3.468	7.741
x2[T.Paris-Est Marne-la-Vallée]					
10.9946	1.171	9.392	0.000	8.700	13.290
x2[T.Paris-Panthéon-Assas]					
13.6818	4.545	3.010	0.003	4.771	22.592
x2[T.Paris-Saclay]					
7.8087	1.172	6.664	0.000	5.512	10.106
x2[T.Pau - Pays de l'Adour]					-
3.7429	1.318	-2.840	0.005	-6.327	-1.159
x2[T.Perpignan - Via Domitia]					
5.1042	1.648	3.098	0.002	1.874	8.334
x2[T.Poitiers]					
1.0710	1.063	1.007	0.314	-1.013	3.155
x2[T.Polytechnique Hauts-de-France]					
9.1594	2.191	4.181	0.000	4.865	13.454
x2[T.Reims - Champagne Ardenne]					
1.1244	1.176	0.956	0.339	-1.180	3.429
x2[T.Rennes 1]					
0.7456	1.070	0.697	0.486	-1.351	2.842
x2[T.Rennes 2]					-
4.0429	1.122	-3.602	0.000	-6.243	-1.843
x2[T.Rouen]					
0.8071	1.070	0.754	0.451	-1.290	2.905
x2[T.Saint-Etienne - Jean Monnet]					
3.5483	1.277	2.779	0.005	1.045	6.052
x2[T.Savoie Mont Blanc]					
4.9508	1.185	4.178	0.000	2.628	7.273
x2[T.Sorbonne Université]					
8.8358	1.147	7.703	0.000	6.587	11.084
x2[T.Strasbourg]					-
0.9084	0.920	-0.988	0.323	-2.712	0.895
x2[T.Toulon]					
12.4681	1.402	8.892	0.000	9.719	15.217
x2[T.Toulouse - Jean Jaures]					-
1.3243	1.159	-1.143	0.253	-3.596	0.947
x2[T.Toulouse 1 - Capitole]					
7.9423	1.162	6.837	0.000	5.665	10.219
x2[T.Toulouse 3 - Paul Sabatier]					
8.9789	1.092	8.220	0.000	6.838	11.120
x2[T.Tours]					-
2.7306	1.128	-2.421	0.016	-4.942	-0.519
x2[T.Université Polytechnique Hauts-de-France]					
6.0134	1.362	4.415	0.000	3.343	8.683
x2[T.Université Sorbonne Paris Nord]					
9.1853	1.572	5.841	0.000	6.103	12.268
x2[T.Université de Paris]					

5.7323	1.248	4.592	0.000	3.285	8.179
x2[T.Versailles Saint-Quentin-en-Yvelines]					
1.7740	1.255	1.414	0.157	-0.685	4.233
x2[T.Évry-Val d'Essonne]					
4.1255	1.363	3.027	0.002	1.454	6.797

=====

Omnibus: 125.517 Durbin-Watson: 1.882

Prob(Omnibus): 0.000 Jarque-Bera (JB): 134.007

Skew: -0.305 Prob(JB): 7.96e-30

Kurtosis: 3.255 Cond. No. 90.0

=====

=====

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

y = df['emplois_stables']
x1 = df['salaire_brut_annuel_estime']
x2 = df['emplois_a_temps_plein']
x3 = df['emplois_cadre']
multivariate_regression = smf.ols(formula=' y ~ x1 + x2 + x3',
data=df).fit()

```

```
print(multivariate_regression.summary())
```

OLS Regression Results

=====

Dep. Variable:	y	R-squared:	0.474
Model:	OLS	Adj. R-squared:	0.474
Method:	Least Squares	F-statistic:	2208.
Date:	Mon, 18 Nov 2024	Prob (F-statistic):	0.00
Time:	16:18:03	Log-Likelihood:	-28179.
No. Observations:	7346	AIC:	5.637e+04
Df Residuals:	7342	BIC:	5.639e+04


```

Df Model:                3
Covariance Type:         nonrobust

=====
=====
              coef      std err          t      P>|t|      [0.025
0.975]
-----
-----
Intercept    -38.9721      1.608     -24.244      0.000     -42.123
-35.821
x1             0.0010     4.64e-05      21.832      0.000      0.001
0.001
x2             0.6924      0.021      33.443      0.000      0.652
0.733
x3             0.2384      0.008      28.249      0.000      0.222
0.255
=====
=====
Omnibus:                215.947    Durbin-Watson:
1.674
Prob(Omnibus):           0.000    Jarque-Bera (JB):
236.079
Skew:                   -0.420    Prob(JB):
5.45e-52
Kurtosis:                3.256    Cond. No.
3.71e+05
=====
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
[2] The condition number is large, 3.71e+05. This might indicate that
there are
strong multicollinearity or other numerical problems.

```

Encodage binaire

```

top5_disciplines = ['Masters enseignement : premier degré', 'Masters
enseignement',
                   'Informatique', 'Gestion']

df['discipline_top5_insertion'] = df['domaine'].apply(lambda x: 1 if x
in top5_disciplines else 0)

df.head()

```

```

{"summary":{"\n  \"name\": \"df\", \n  \"rows\": 7346, \n  \"fields\": [\n    {\n      \"column\": \"annee\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 2, \n        \"min\": 2011, \n        \"max\": 2020, \n        \"num_unique_values\": 10, \n        \"samples\": [\n          2011, \n          2018, \n          2014\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"diplome\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 2, \n        \"samples\": [\n          \"MASTER ENS\", \n          \"MASTER LMD\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"etablissement\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 83, \n        \"samples\": [\n          \"Grenoble Alpes\", \n          \"Paris 3 - Sorbonne Nouvelle\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"academie\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 26, \n        \"samples\": [\n          \"Aix-Marseille\", \n          \"Lyon\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"domaine\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 5, \n        \"samples\": [\n          \"Masters enseignement\", \n          \"Sciences humaines et sociales\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"discipline\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 20, \n        \"samples\": [\n          \"Lettres, langues, arts\", \n          \"Droit\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"situation\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 2, \n        \"samples\": [\n          \"30 mois apr\\u00e8s le dipl\\u00f4me\", \n          \"18 mois apr\\u00e8s le dipl\\u00f4me\"\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"poids_de_la_discipline\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 25.76148266938614, \n        \"min\": 2.0, \n        \"max\": 100.0, \n        \"num_unique_values\": 94, \n        \"samples\": [\n          62.0, \n          42.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"taux_dinsertion\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 5.806050159948711, \n        \"min\": 63.0, \n        \"max\": 100.0, \n        \"num_unique_values\": 37, \n        \"samples\": [\n          89.0, \n          78.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }\n    }, \n    {\n      \"column\": \"emplois_cadre_ou_professions_intermediaires\", \n

```

```

\"properties\": {\n      \"dtype\": \"number\", \n      \"std\": 10.12488815564198, \n      \"min\": 3.0, \n      \"max\": 100.0, \n      \"num_unique_values\": 66, \n      \"samples\": [\n        48.0, \n        32.0\n      ], \n      \"semantic_type\": \"\", \n      \"description\": \"\" \n    }, \n    {\n      \"column\": \"emplois_stables\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 15.464611385219834, \n        \"min\": 19.0, \n        \"max\": 100.0, \n        \"num_unique_values\": 79, \n        \"samples\": [\n          84.0, \n          58.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      }, \n      {\n        \"column\": \"emplois_a_temps_plein\", \n        \"properties\": {\n          \"dtype\": \"number\", \n          \"std\": 7.734538748728703, \n          \"min\": 45.0, \n          \"max\": 100.0, \n          \"num_unique_values\": 52, \n          \"samples\": [\n            92.0, \n            68.0\n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"salaire_net_median_des_emplois_a_temps_plein\", \n          \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 256.64650646796997, \n            \"min\": 1280.0, \n            \"max\": 3150.0, \n            \"num_unique_values\": 154, \n            \"samples\": [\n              1750.0, \n              2770.0\n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          }, \n          {\n            \"column\": \"salaire_brut_annuel_estime\", \n            \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": 4003.3250718655077, \n              \"min\": 20000.0, \n              \"max\": 49100.0, \n              \"num_unique_values\": 223, \n              \"samples\": [\n                26500.0, \n                28500.0\n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n            }, \n            {\n              \"column\": \"de_diplomes_boursiers\", \n              \"properties\": {\n                \"dtype\": \"number\", \n                \"std\": 8.57587581394852, \n                \"min\": 0.0, \n                \"max\": 64.0, \n                \"num_unique_values\": 147, \n                \"samples\": [\n                  30.78, \n                  30.34\n                ], \n                \"semantic_type\": \"\", \n                \"description\": \"\" \n              }, \n              {\n                \"column\": \"salaire_net_mensuel_median_regional\", \n                \"properties\": {\n                  \"dtype\": \"number\", \n                  \"std\": 136.30629006623818, \n                  \"min\": 1580.0, \n                  \"max\": 2300.0, \n                  \"num_unique_values\": 44, \n                  \"samples\": [\n                    1990.0, \n                    1750.0\n                  ], \n                  \"semantic_type\": \"\", \n                  \"description\": \"\" \n                }, \n                {\n                  \"column\": \"emplois_cadre\", \n                  \"properties\": {\n                    \"dtype\": \"number\", \n                    \"std\": 18.575640312995823, \n                    \"min\": 1.0, \n                    \"max\": 100.0, \n                    \"num_unique_values\": 94, \n                    \"samples\": [\n                      45.0, \n                      74.0\n                    ], \n                    \"semantic_type\": \"\", \n                    \"description\": \"\" \n                  }, \n                  {\n                    \"column\": \"emplois_exterieurs_a_la_region_de_luniversite\", \n                    \"properties\": {\n                      \"dtype\": \"number\", \n                      \"std\": 17.9438247467822, \n                      \"min\": 2.0, \n                      \"max\": 100.0, \n                      \"num_unique_values\": 95, \n                      \"samples\": [\n                        76.0, \n

```

```

69.0\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\":\n          \"femmes\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 19.176047452353345,\n            \"min\": 2.0,\n            \"max\": 98.0,\n            \"num_unique_values\": 96,\n            \"samples\": [\n              23.0,\n              41.0\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          },\n          {\n            \"column\":\n            \"discipline_top5_insertion\",\n            \"properties\": {\n              \"dtype\": \"number\",\n              \"std\": 0,\n              \"min\": 0,\n              \"max\": 1,\n              \"num_unique_values\": 2,\n              \"samples\": [\n                1,\n                0\n              ],\n              \"semantic_type\":\n              \"\",\n              \"description\": \"\"\n            }\n          }\n        }\n      ],\n      \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```

df['Etablissement_parisien'] = df['academie'].apply(lambda x: 1 if x
== 'Paris' else 0)

```

```
df.head()
```

```
{"type": "dataframe", "variable_name": "df"}
```

```

y = df['emplois_stables']
x1 = df['salaire_brut_annuel_estime']
x2 = df['emplois_a_temps_plein']
x3 = df['emplois_cadre']
x4 = df['Etablissement_parisien']
x5 = df['discipline_top5_insertion']
multivariate_regression = smf.ols(formula=' y ~  x1 + x2 + x3 + x4 +
x5', data=df).fit()

```

```
print(multivariate_regression.summary())
```

OLS Regression Results

```

=====
=====
Dep. Variable:          y      R-squared:
0.592
Model:                OLS      Adj. R-squared:
0.592
Method:               Least Squares      F-statistic:
2132.
Date:                 Mon, 18 Nov 2024      Prob (F-statistic):
0.00
Time:                 16:18:04      Log-Likelihood:
-27246.
No. Observations:      7346      AIC:
5.450e+04
Df Residuals:          7340      BIC:
5.454e+04

```

Df Model: 5

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025
0.975]					

Intercept	-39.8129	1.418	-28.075	0.000	-42.593
-37.033					
x1	0.0020	4.66e-05	43.825	0.000	0.002
0.002					
x2	0.4814	0.019	25.479	0.000	0.444
0.518					
x3	0.0661	0.009	7.658	0.000	0.049
0.083					
x4	-8.0533	0.424	-18.999	0.000	-8.884
-7.222					
x5	20.2911	0.490	41.410	0.000	19.331
21.252					
=====					
=====					
Omnibus:	186.177	Durbin-Watson:			
1.725					
Prob(Omnibus):	0.000	Jarque-Bera (JB):			
212.817					
Skew:	-0.354	Prob(JB):			
6.13e-47					
Kurtosis:	3.439	Cond. No.			
3.71e+05					
=====					
=====					

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.71e+05. This might indicate that there are strong multicollinearity or other numerical problems.

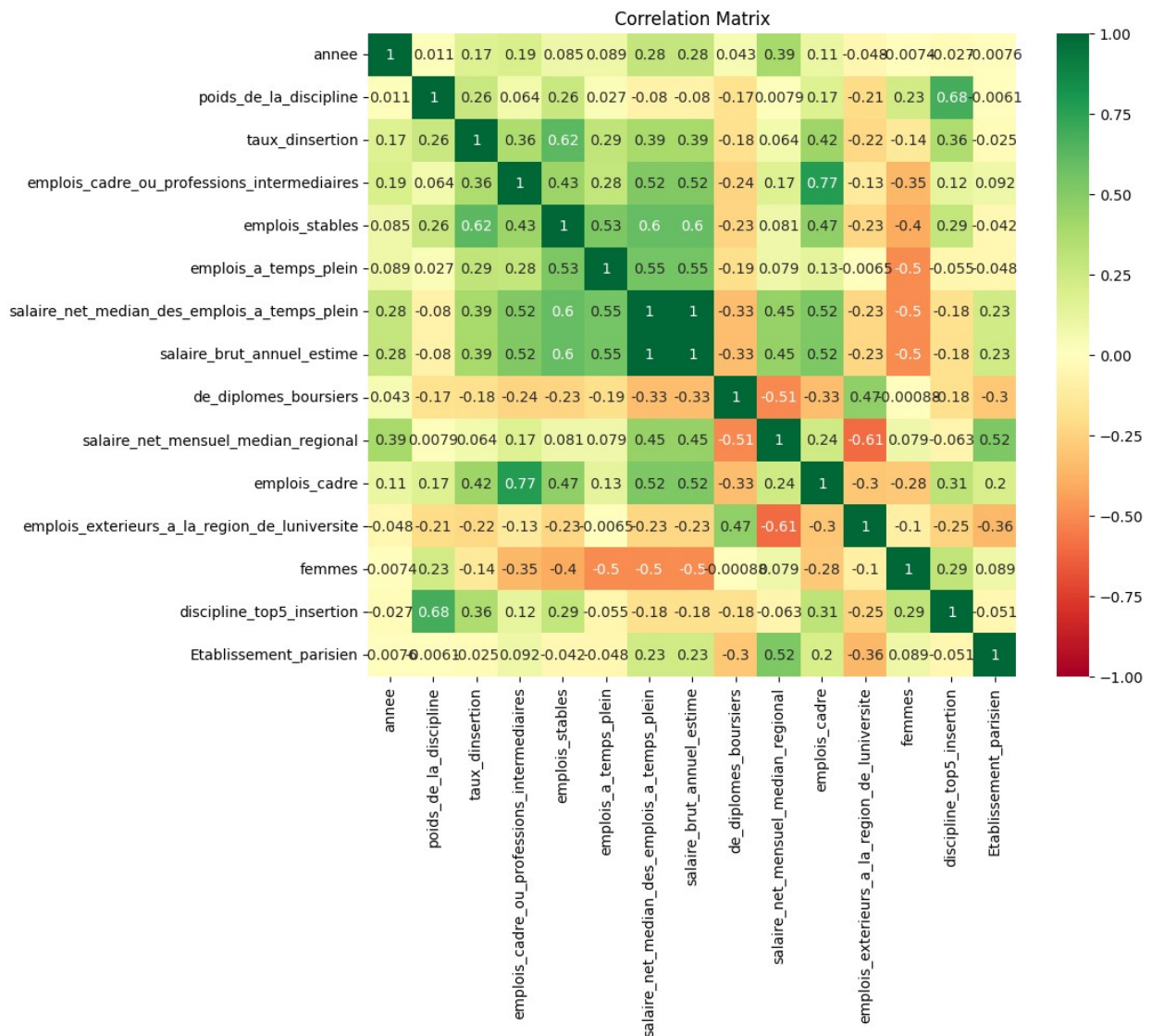
df

```
{"type": "dataframe", "variable_name": "df"}
```

```
numeric_df = df.select_dtypes(include=[np.number])
```

```
correlation_matrix = numeric_df.corr()
```

```
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="RdYlGn", vmin=-1,
vmax=1)
plt.title("Correlation Matrix")
plt.show()
```



```
df.shape
(7346, 21)
df.to_csv('/content/df_econometrics.csv', index=False)
shutil.move('/content/df_econometrics.csv',
'/content/drive/MyDrive/df_econometrics.csv')
{"type": "string"}
```